

Learning goals

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5. Explain the causes of replicative stress

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6. Explain how homologous recombination underpins DNA replication

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7. Describe the roles of important common factors in homologous recombination and replication stress response

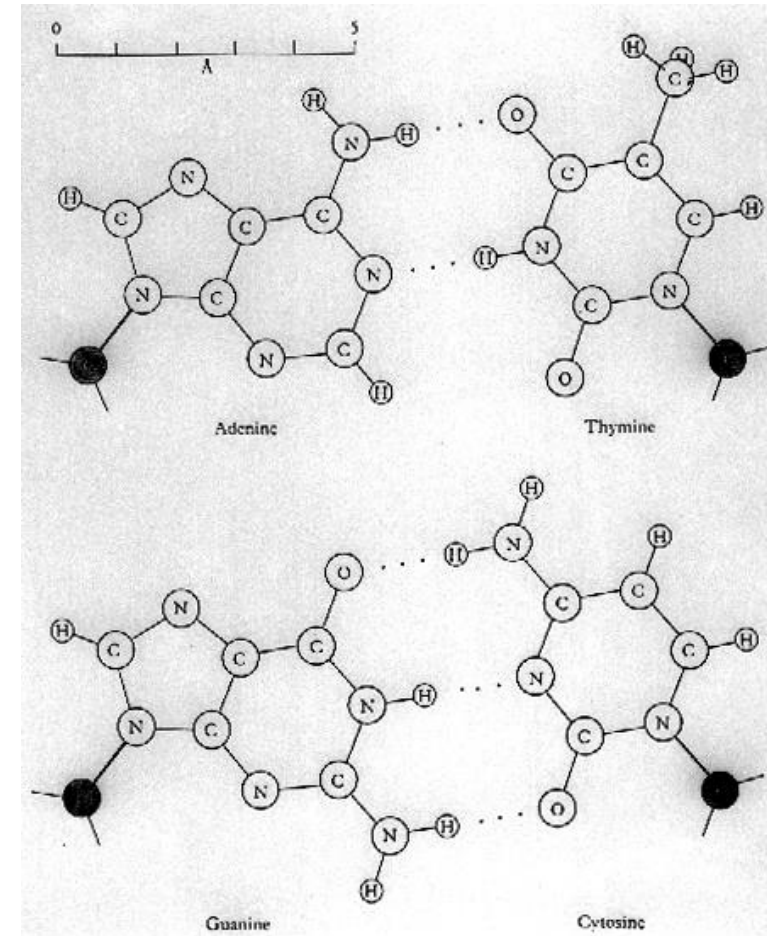
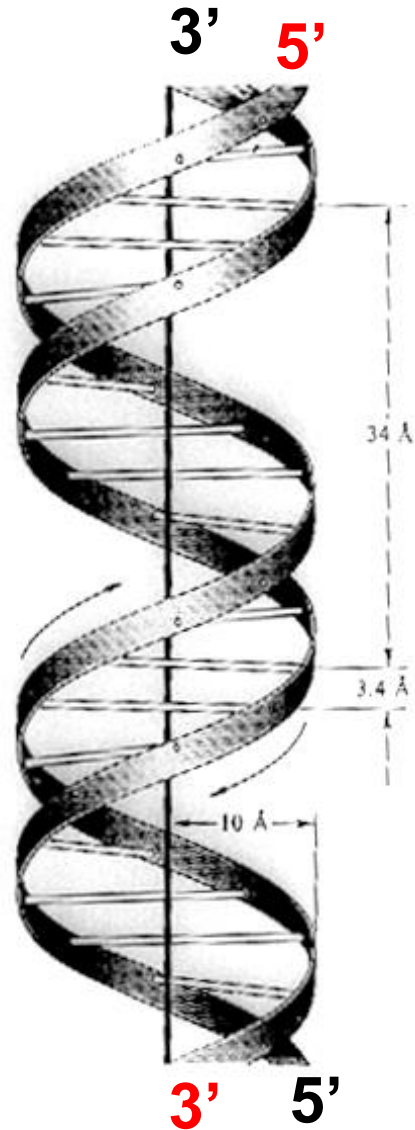
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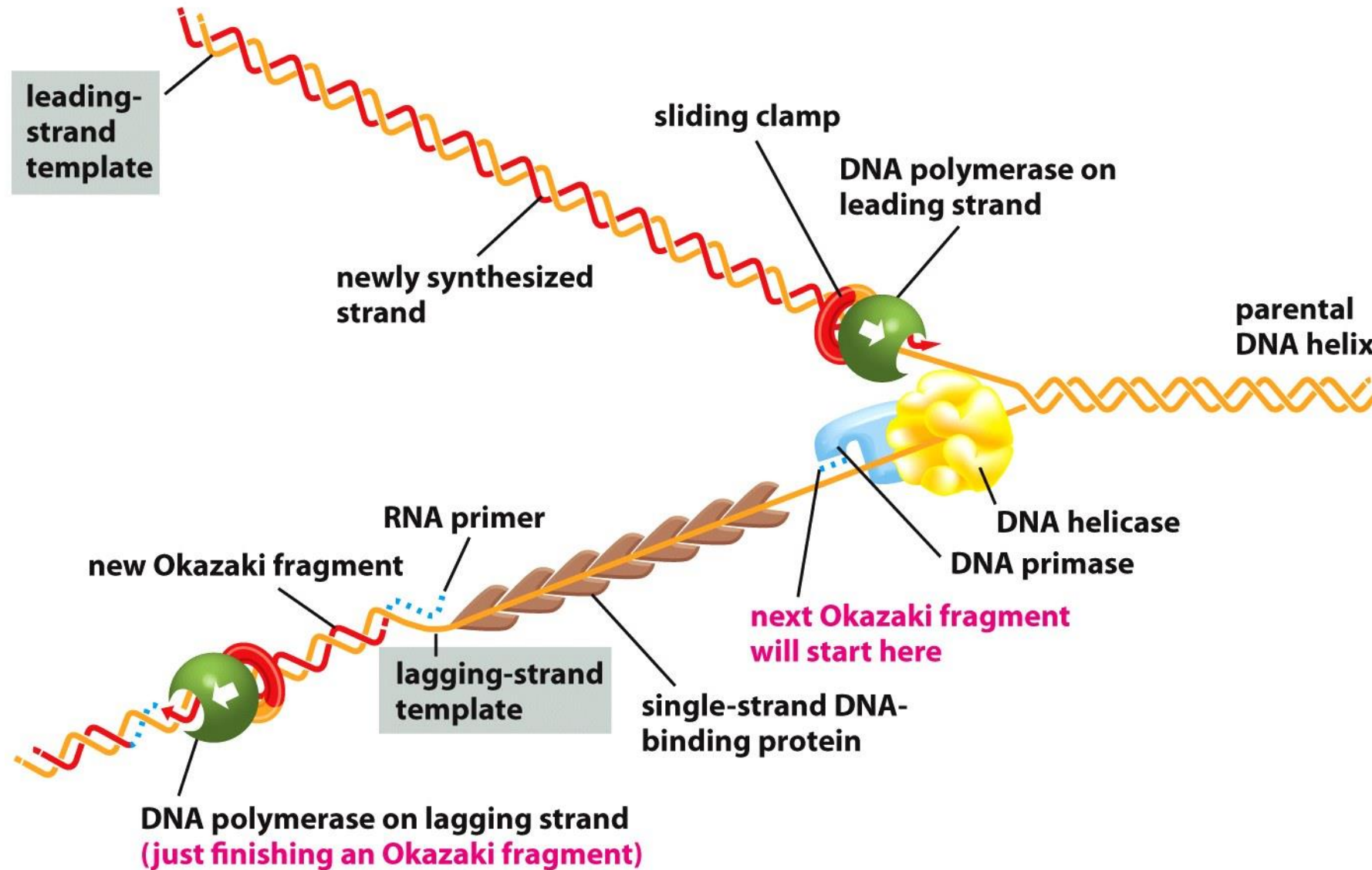
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7. Describe the roles of important common factors in homologous recombination and replication stress response
8. Describe the main techniques used to study homologous recombination and replication stress
9. Explain how replication stress and homologous recombination defects can be exploited for cancer treatments

The structure of DNA



Anatomy of a replication fork



Anatomy of a replication fork

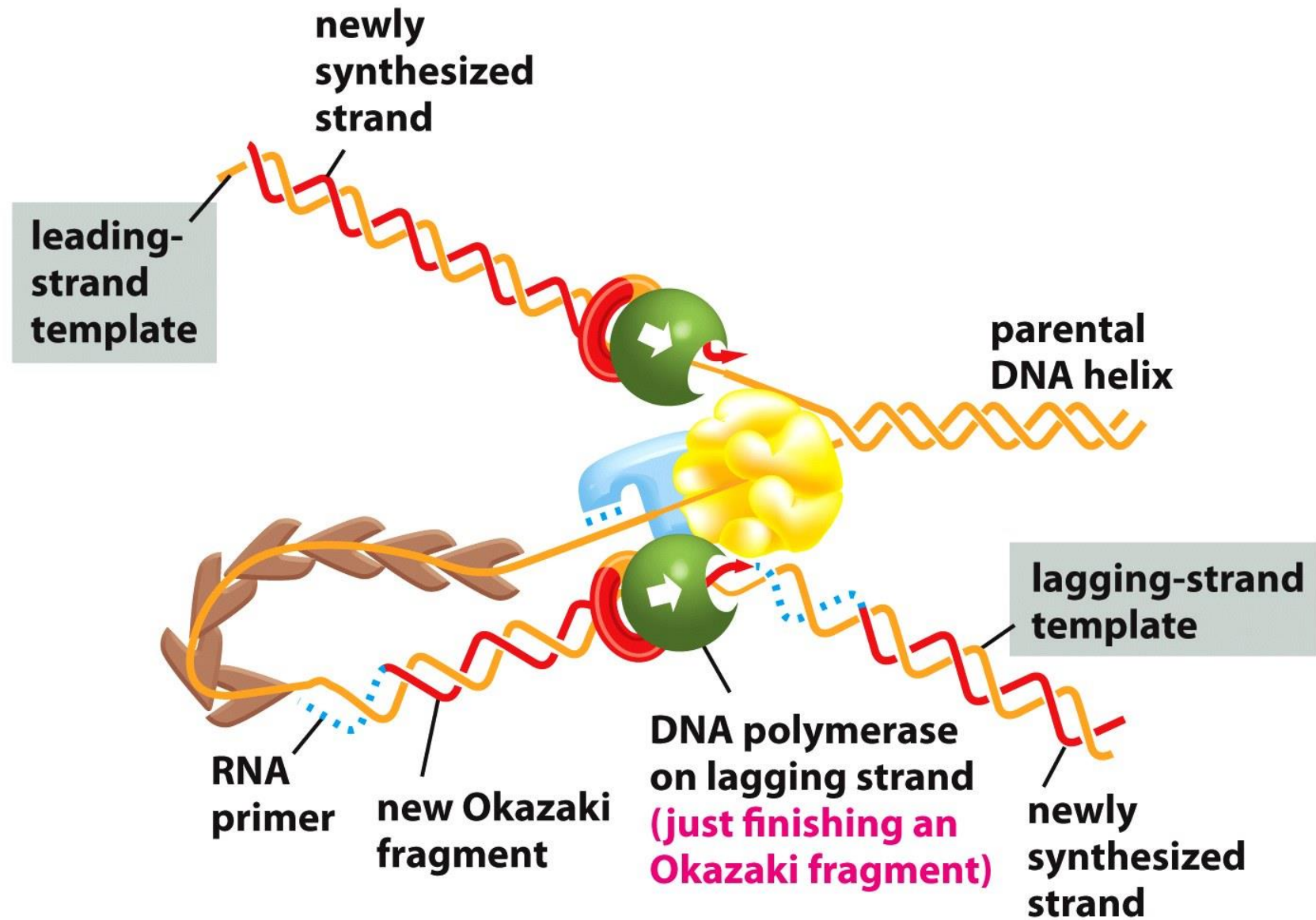
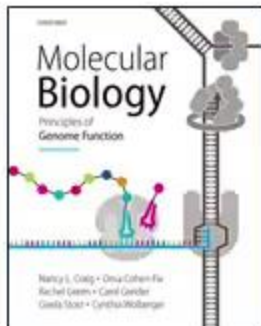


Figure 6-17b *Essential Cell Biology* (© Garland Science 2010)



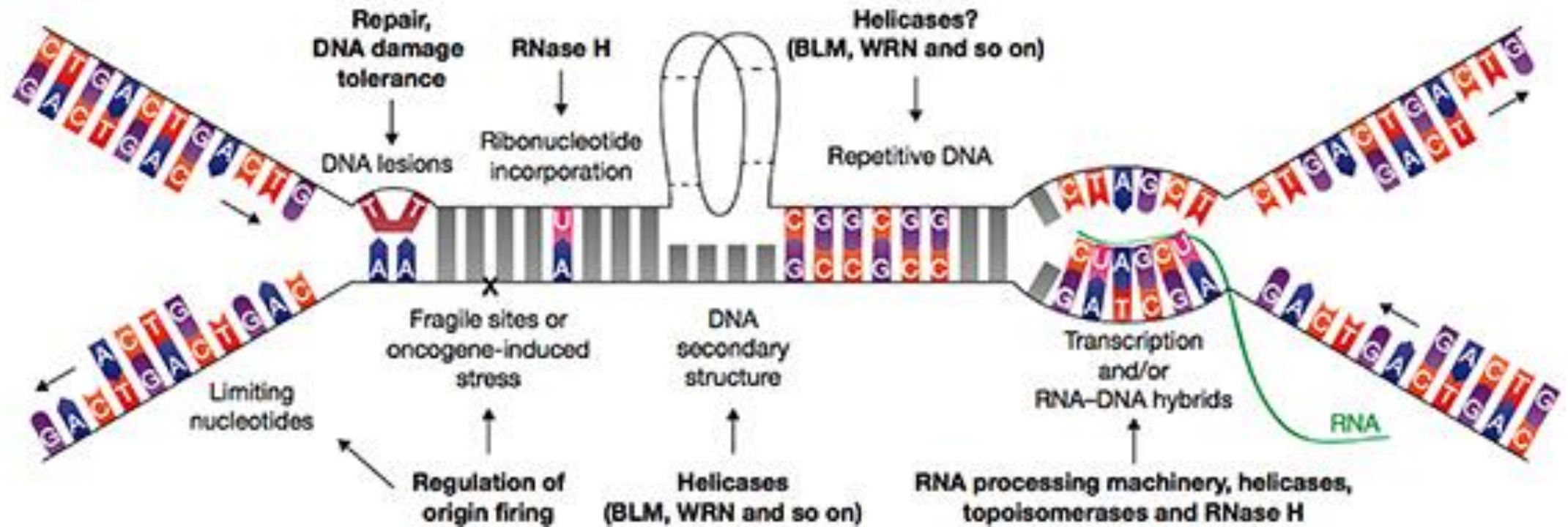
Molecular Biology: Principles of Genome Function

Second Edition

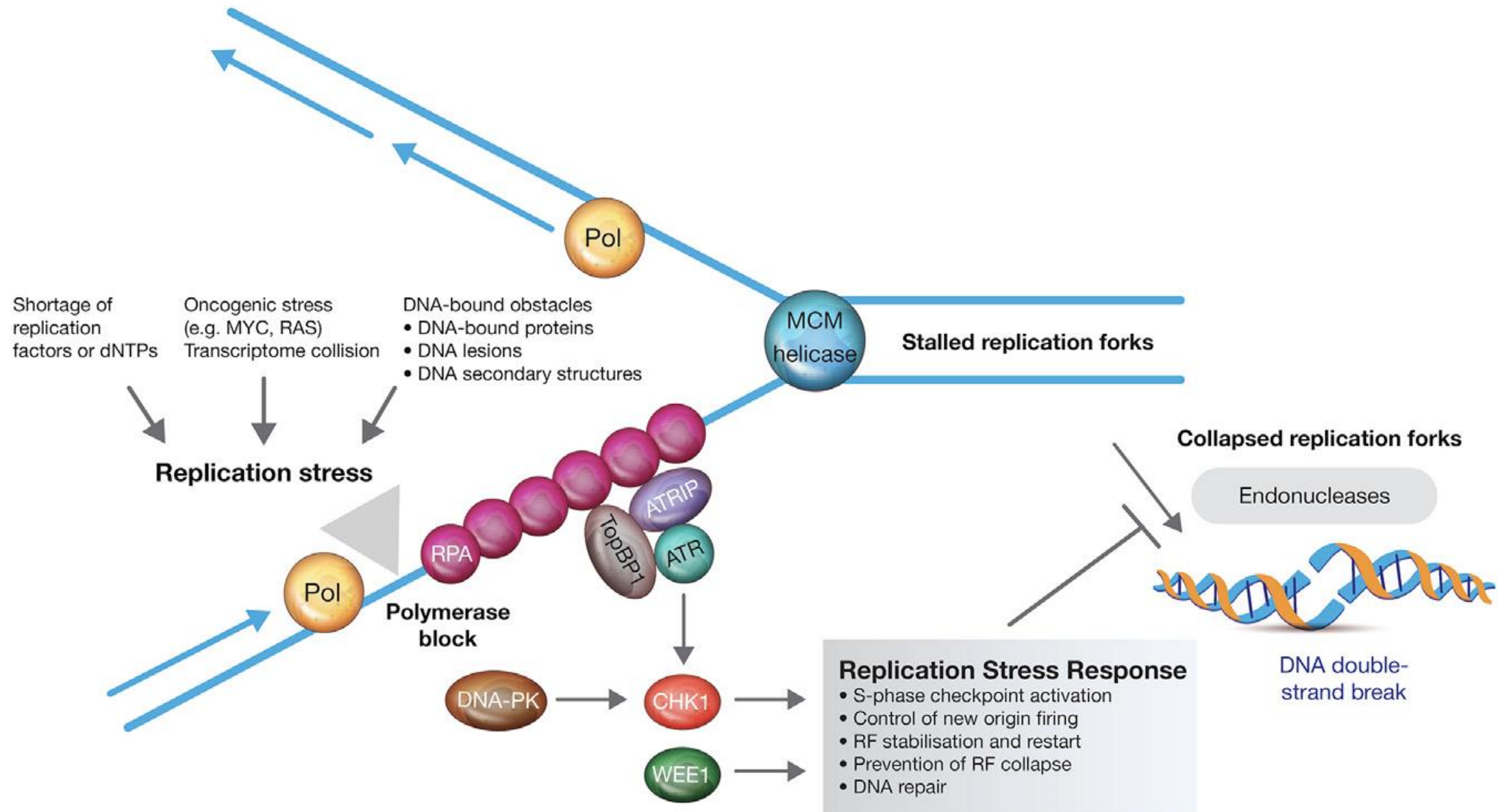
OXFORD
UNIVERSITY PRESS

Animation 2: Replication fork coupling

DNA replication stress and its causes

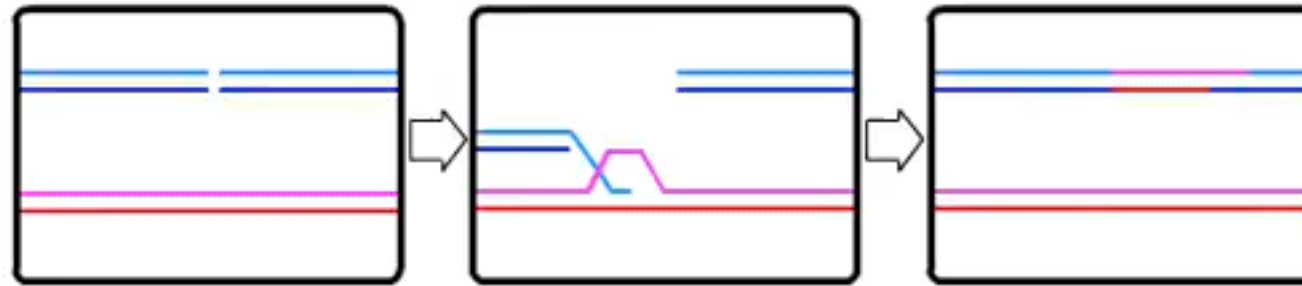


Consequences of DNA replication stress



How to repair DSBs during S- Phase

Homologous Recombination Repair
of a Two-Ended Double Strand Break
(Synthesis-Dependent Strand Annealing)



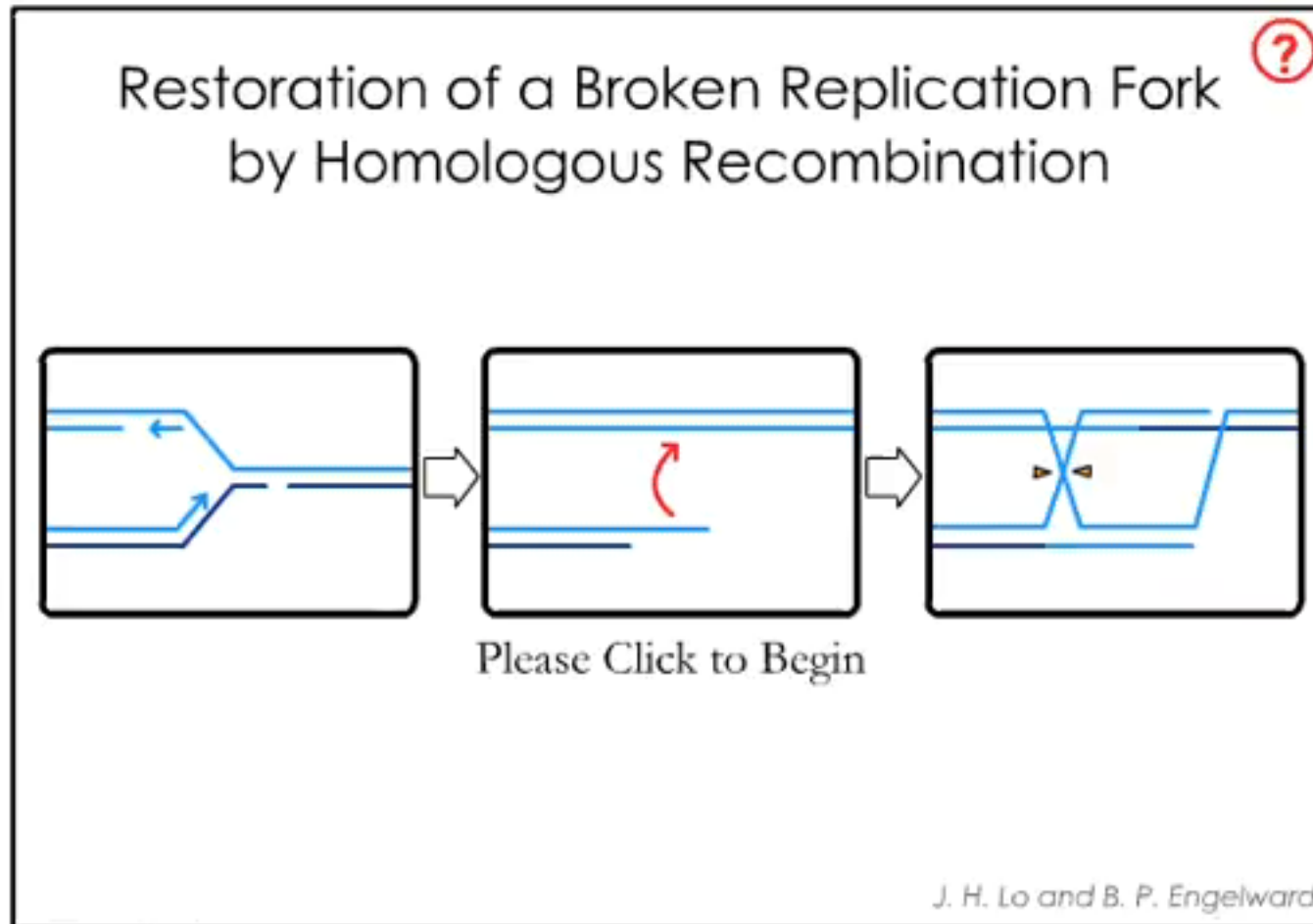
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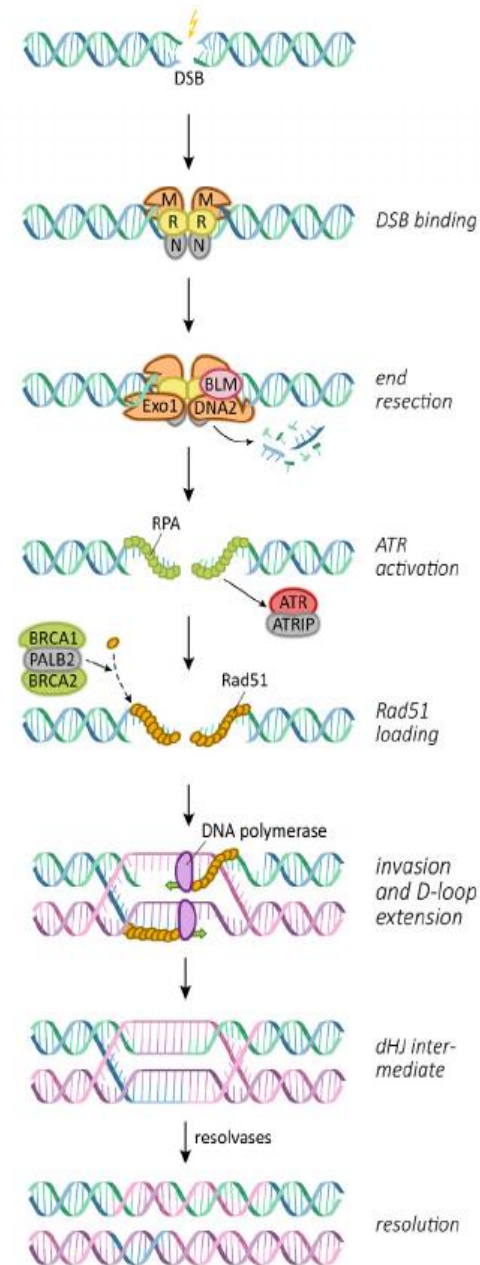
J. H. Lo and B. P. Engelward

**Two-Ended Double Strand Break
Repair via Synthesis dependent
Strand Annealing(SDSA)**

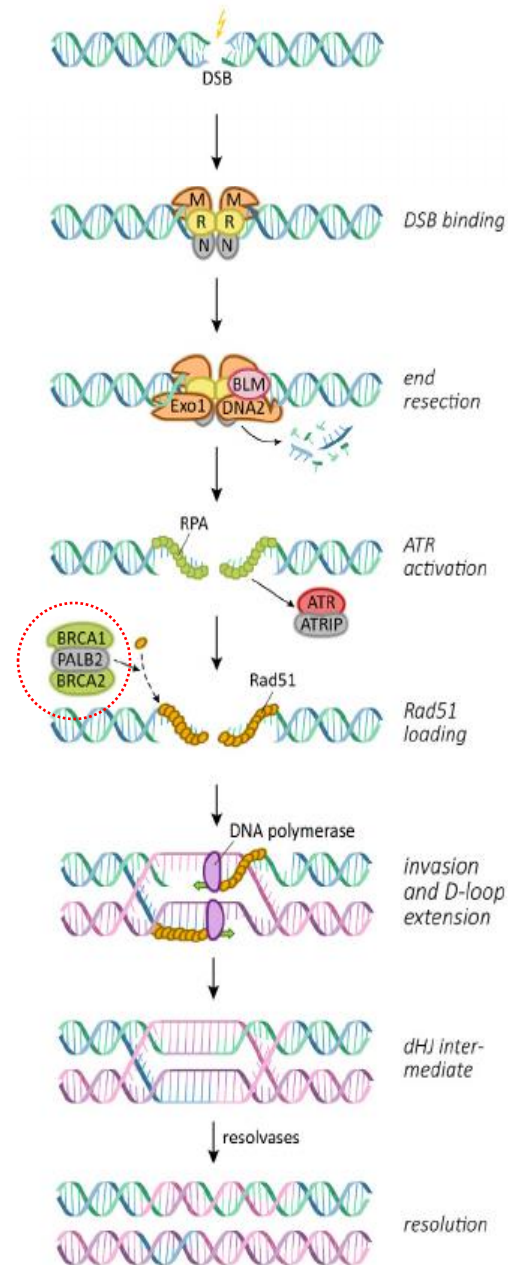
Replication associated double-strand break repair by Homologous recombination



Factors involved in Homologous recombination



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What do we know about BRCA proteins

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- BRCA1/2 mutations **promote** tumor growth.

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- BRCA1/2 mutations **promote** tumor growth.
- BRCA1/2 deletion in primary cells **inhibits** growth.

What do we know about BRCA proteins

- BRCA1/2 mutations **promote** tumor growth.
- BRCA1/2 deletion in primary cells **inhibits** growth.
- Loss of homologous recombination is believed to be the **major catalyst** for tumor promotion and growth inhibition.

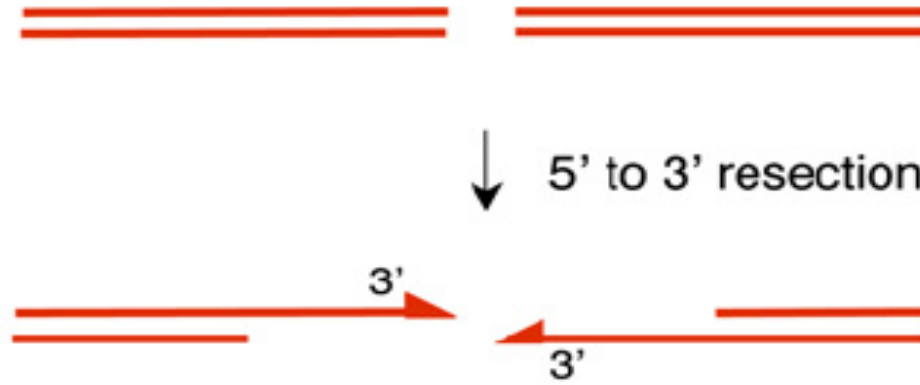
BRCA involved in DSB repair

Homologous recombination



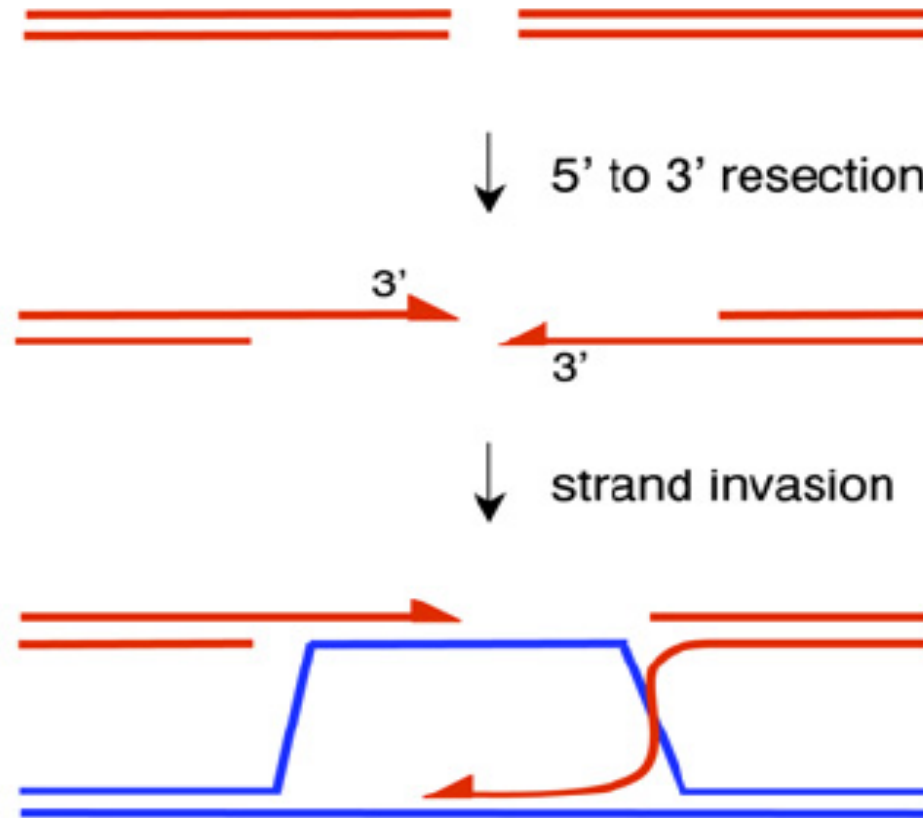
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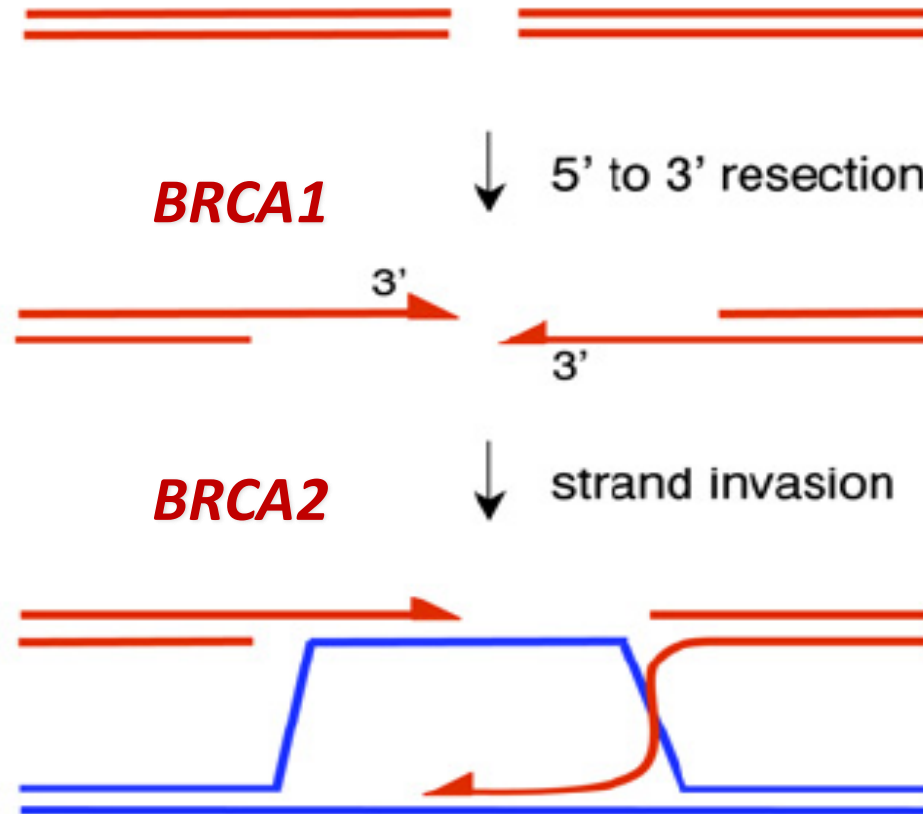
BRCA involved in DSB repair

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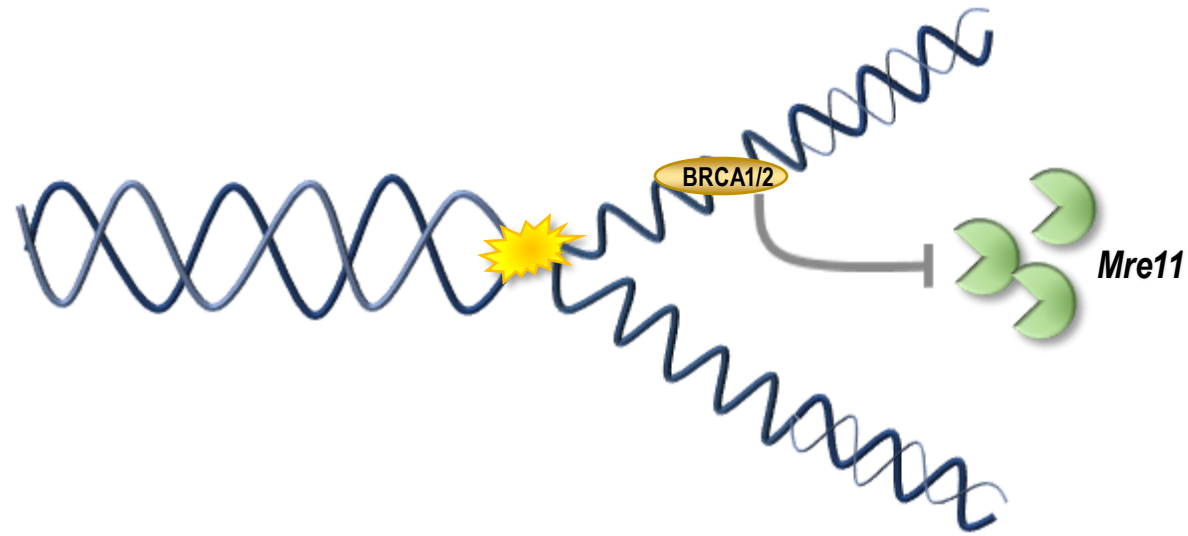


BRCA involved in DSB repair

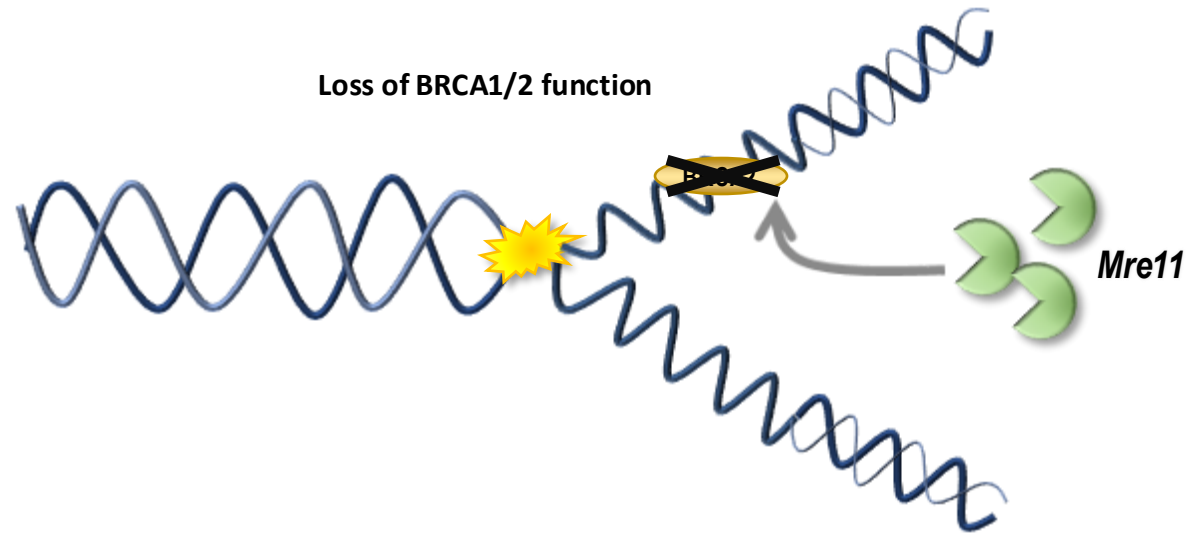
Homologous recombination



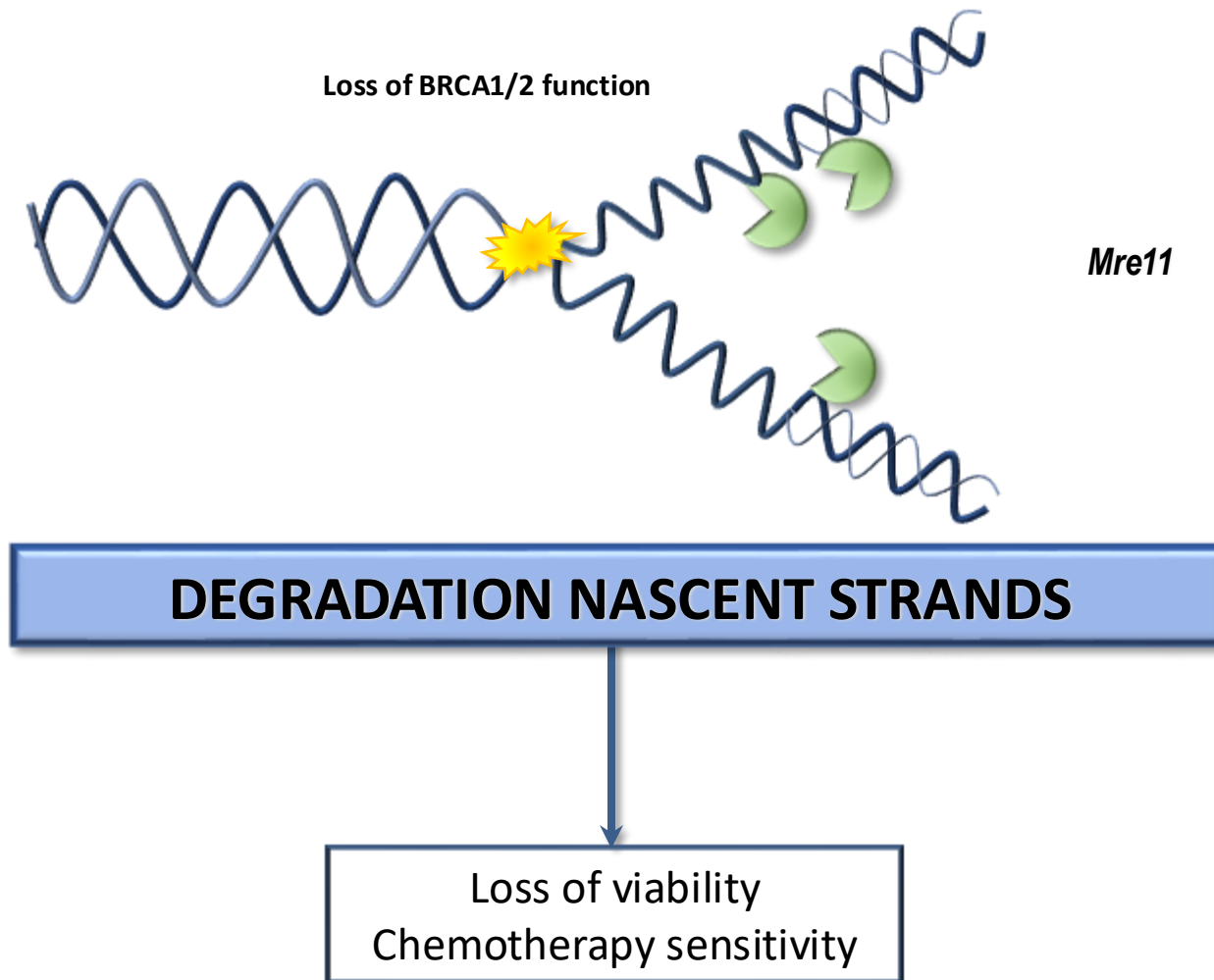
BRCA1/2 protects replication forks from nuclease-mediated degradation



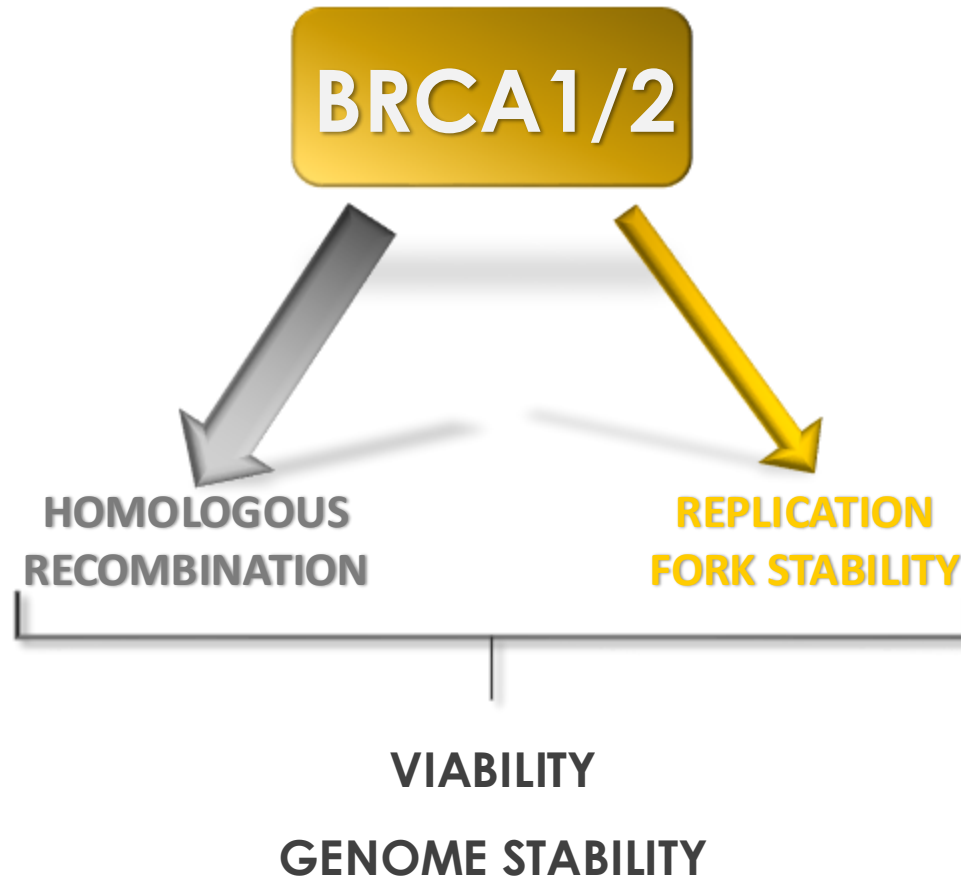
BRCA1/2 protects replication forks from nuclease-mediated degradation



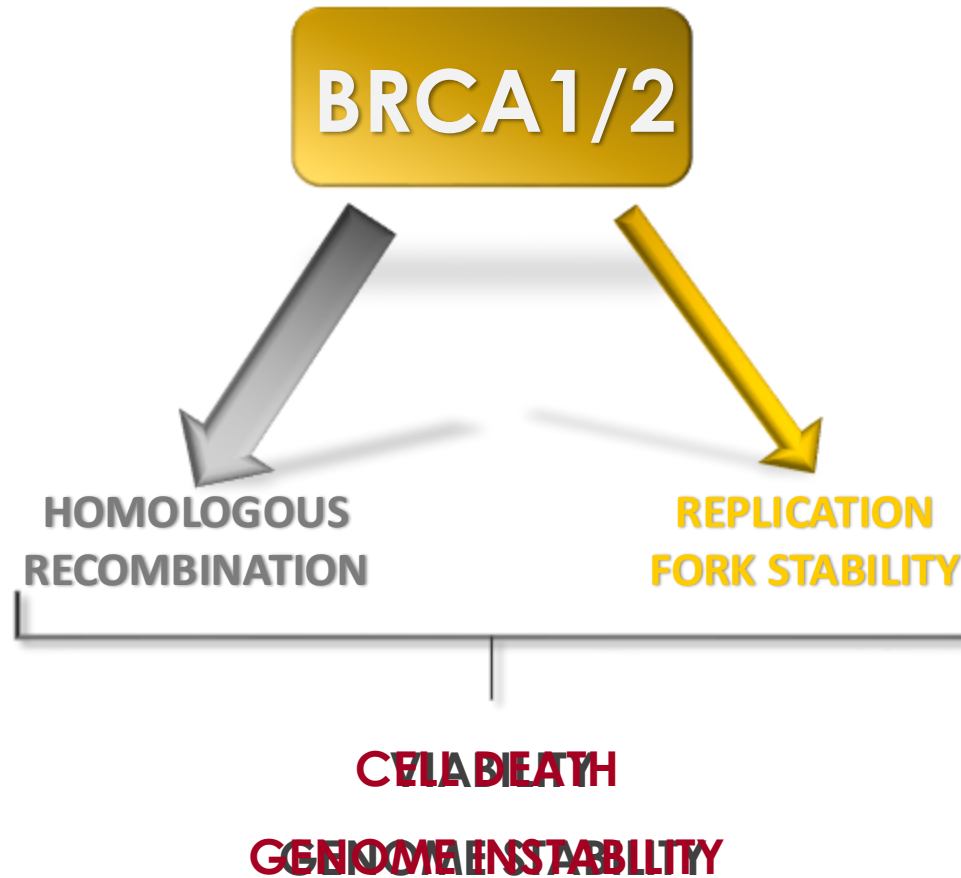
BRCA1/2 protects replication forks from nuclease-mediated degradation



Multiple functions of BRCA1/2



Multiple functions of BRCA1/2



Techniques used to study DNA replication



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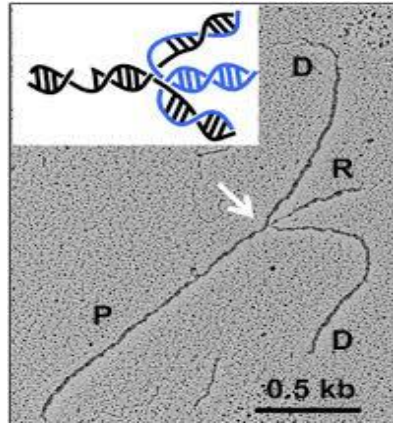
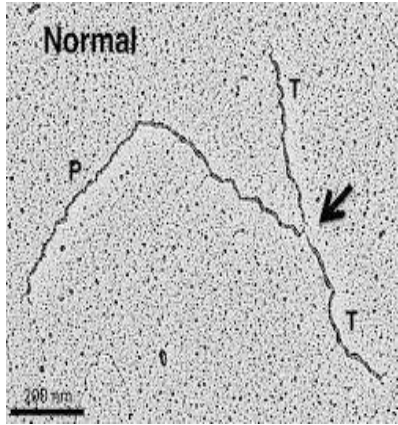
Electron Microscopy



Techniques used to study DNA replication

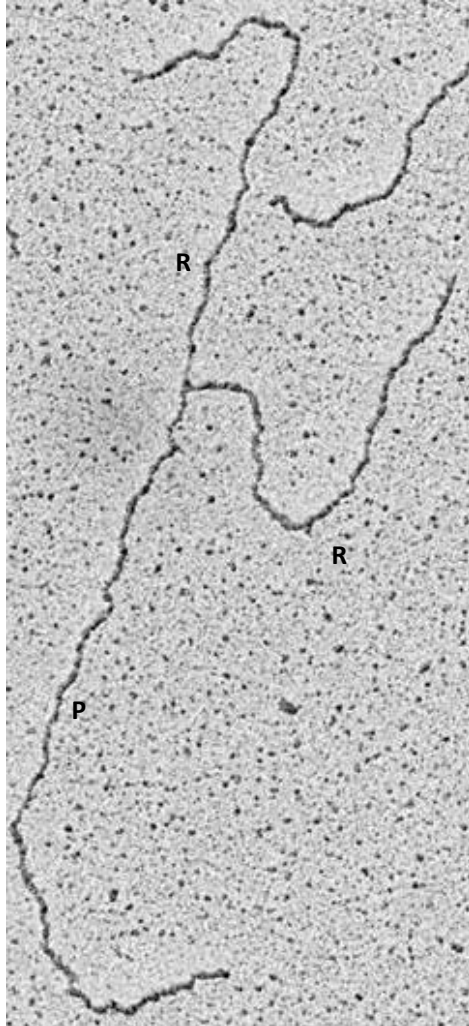


Electron Microscopy

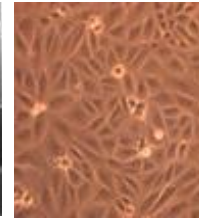
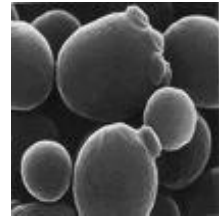


Structural visualization
of replication
intermediates

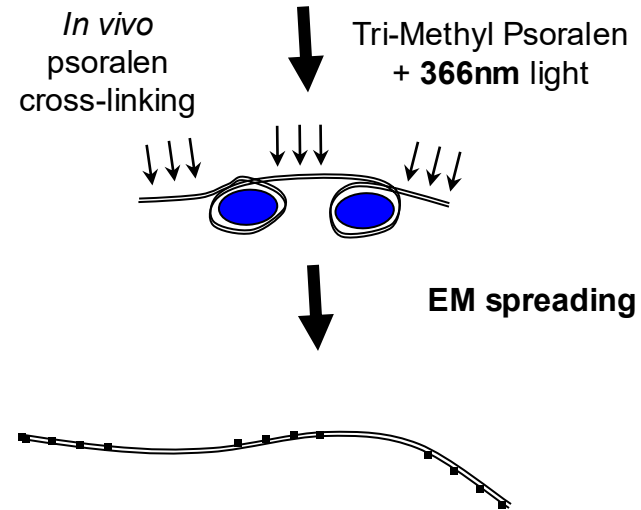
In vivo investigation into DNA replication intermediates



R. Zelweger and M. Lopes, Meth Mol Biol 2017



J. Sogo, ETH Zürich



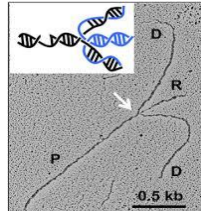
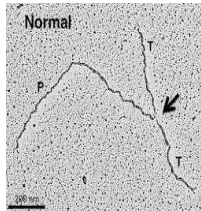
Stabilization of *in vivo* replication intermediates

Techniques used to study DNA replication

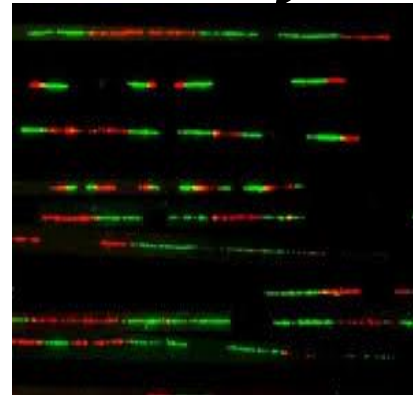


Electron Microscopy

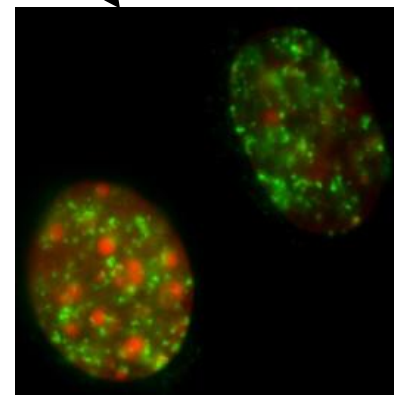
Cell Biology



Structural visualization of replication intermediates

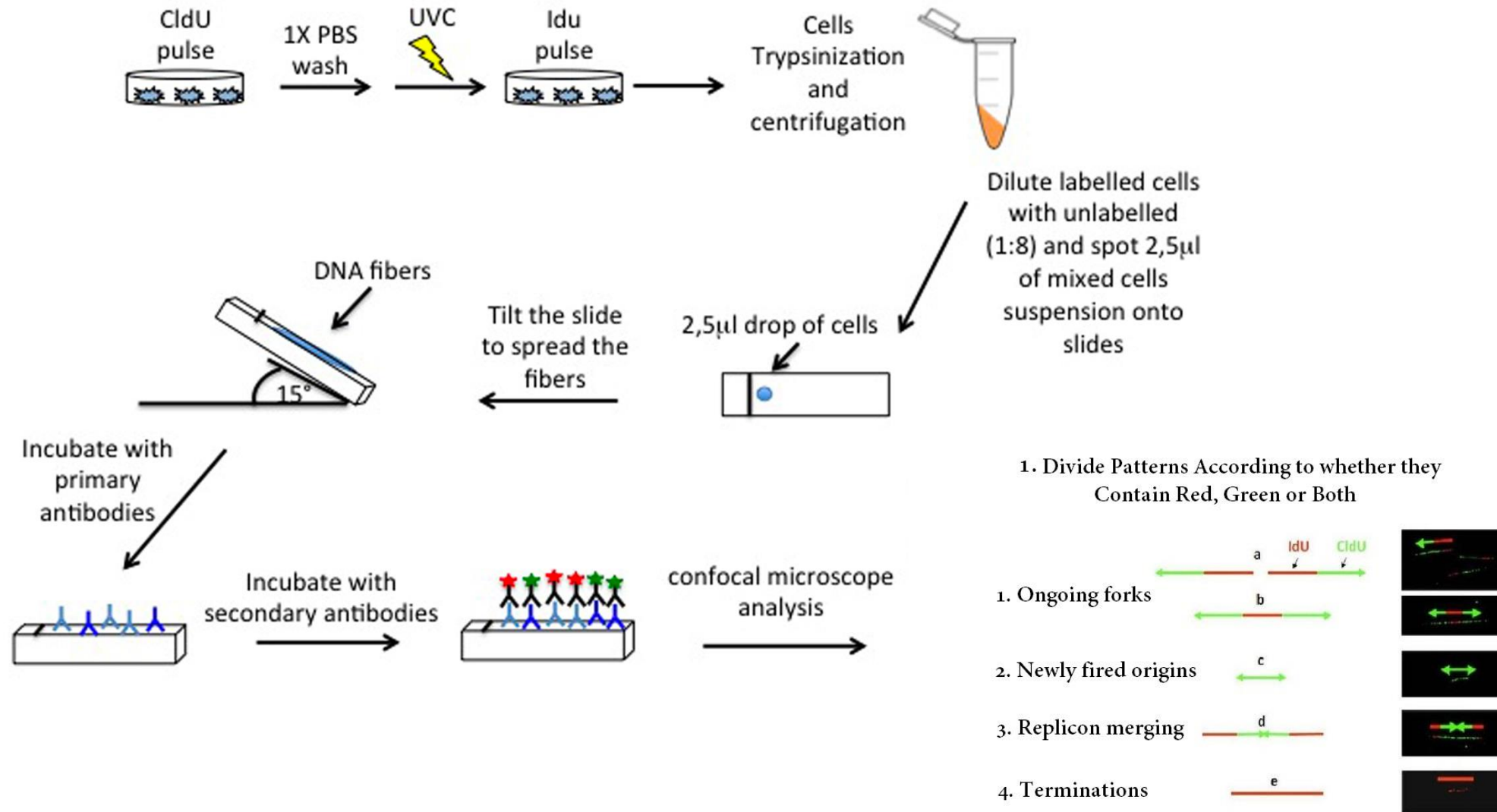


Single molecule analysis of DNA replication



Immunofluorescence analysis of replication proteins

DNA fibers to study DNA replication dynamics

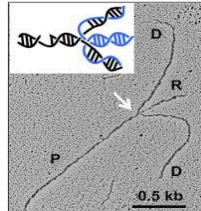
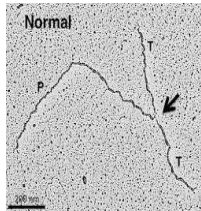


Techniques used to study DNA replication

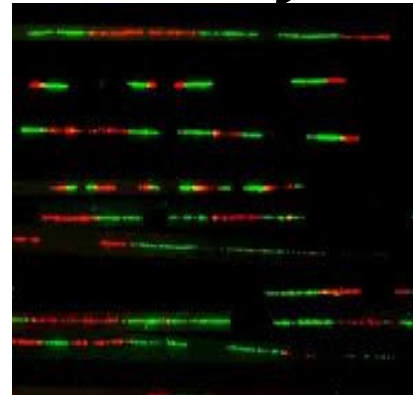


Electron Microscopy

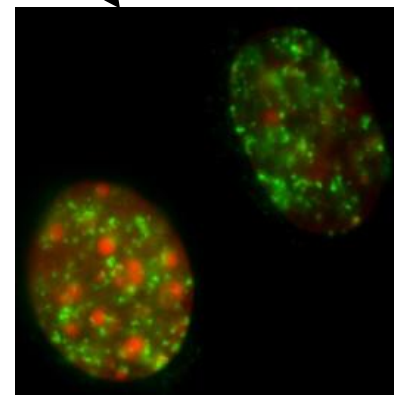
Cell Biology



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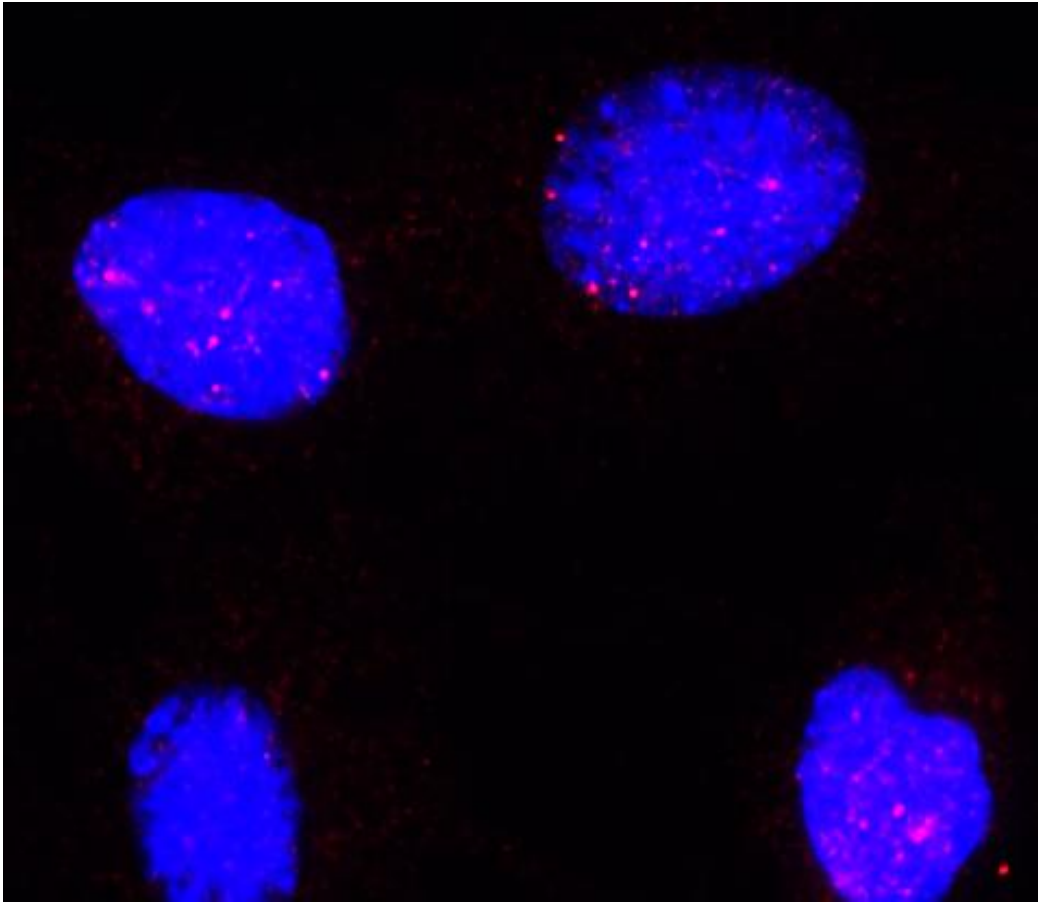
Single molecule analysis of DNA replication



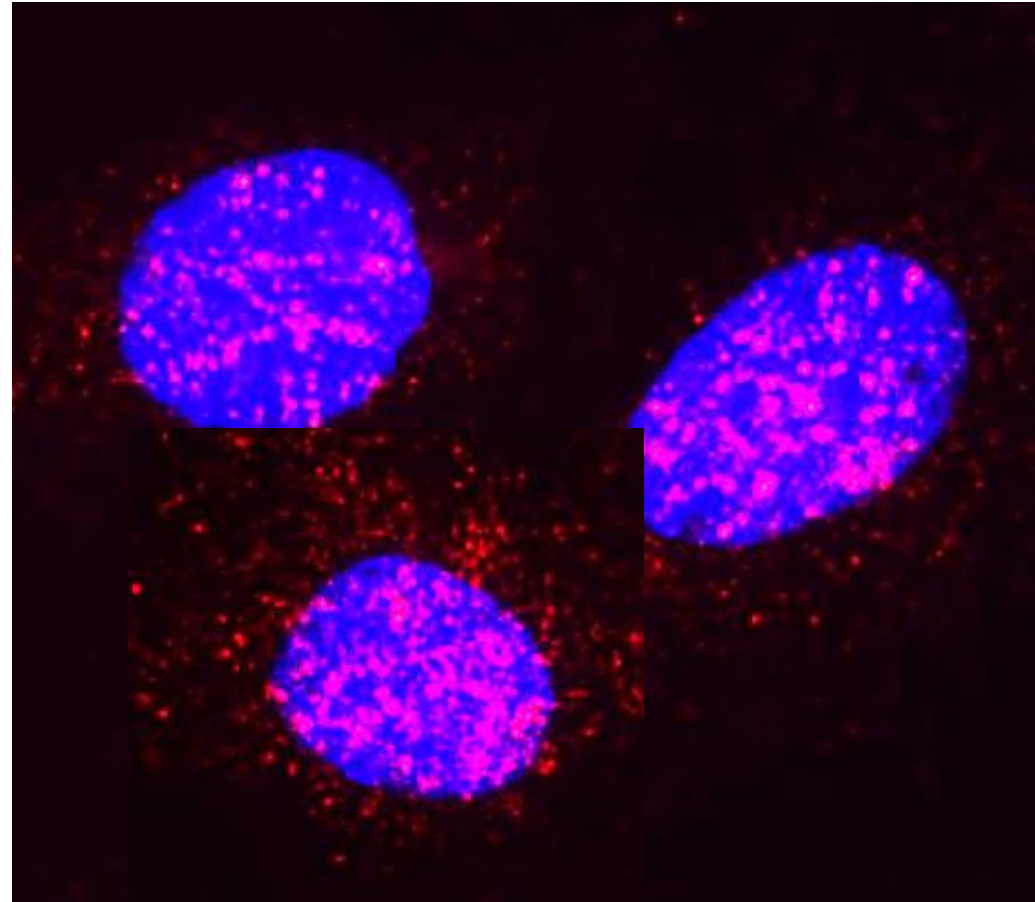
Immunofluorescence analysis of replication proteins

Protein localization at replication forks

Untreated



Replication stress



RAD51

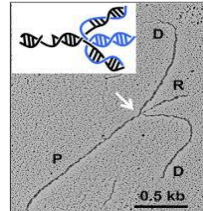
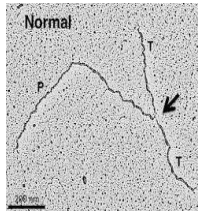
Techniques used to study DNA replication



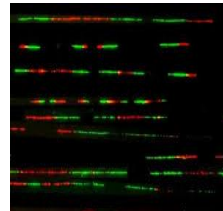
Electron Microscopy

Cell Biology

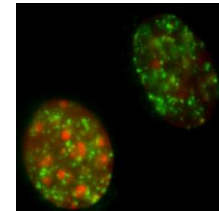
Biochemistry and Proteomics



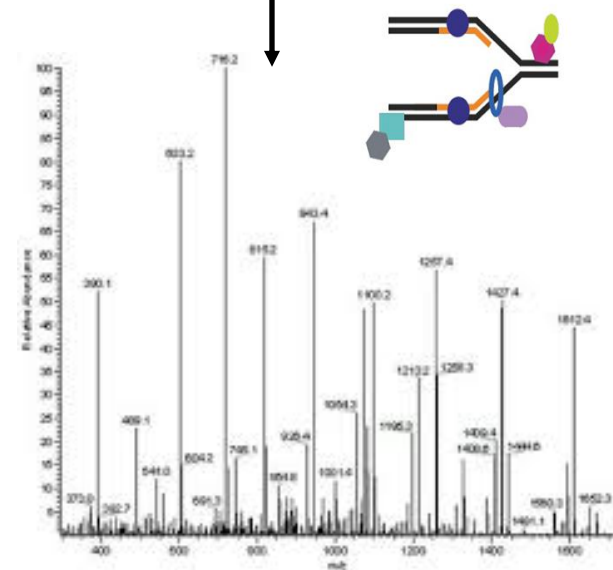
Structural visualization of replication intermediates



Single molecule analysis of DNA replication

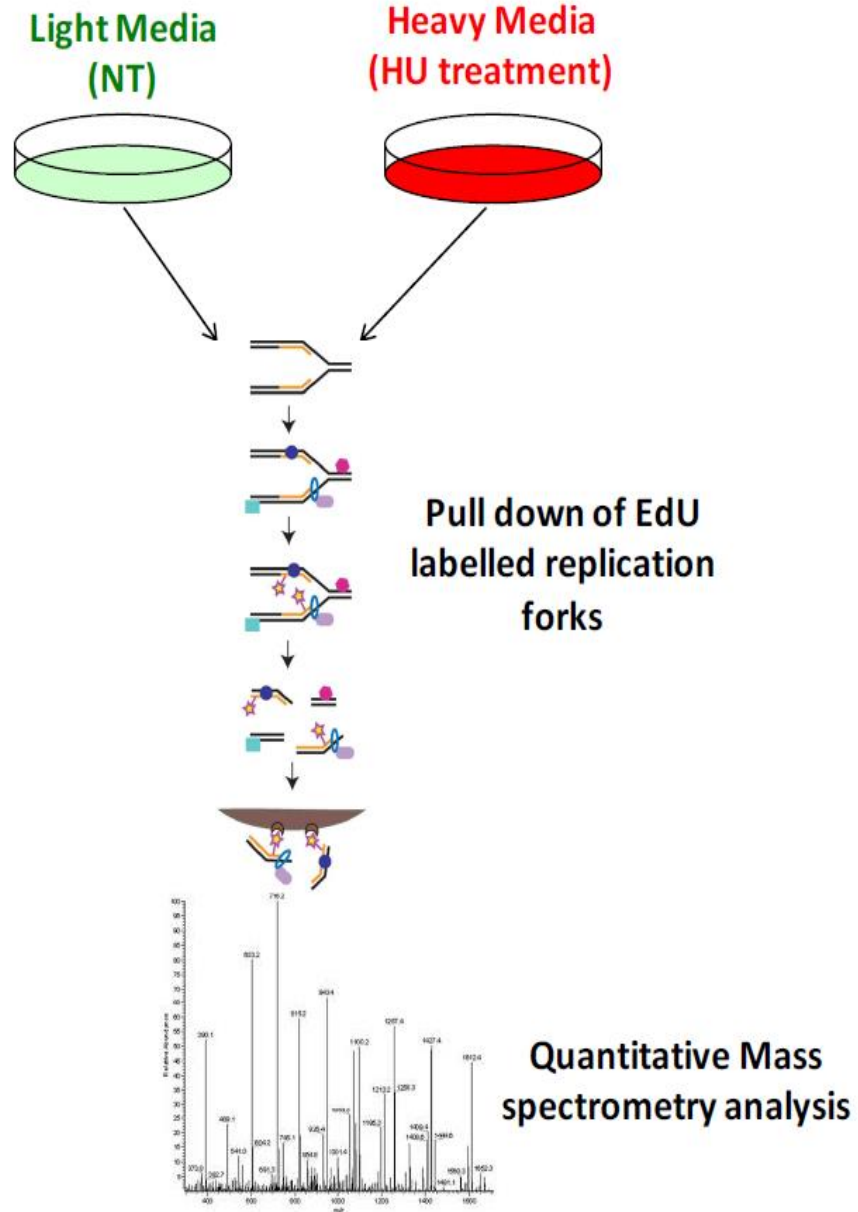


Immunofluorescence analysis of replication proteins



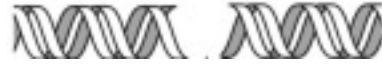
Analysis of protein dynamics at replication forks

Proteomes at replication forks



How to study HR?

Double-strand breaks
(IR)



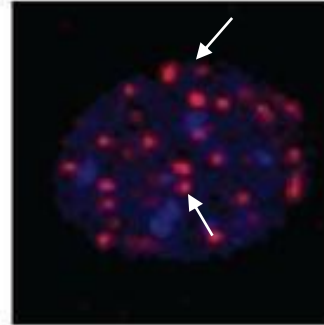
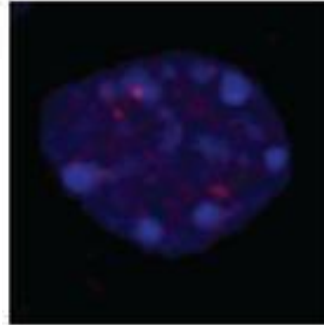
RAD51

IR

–

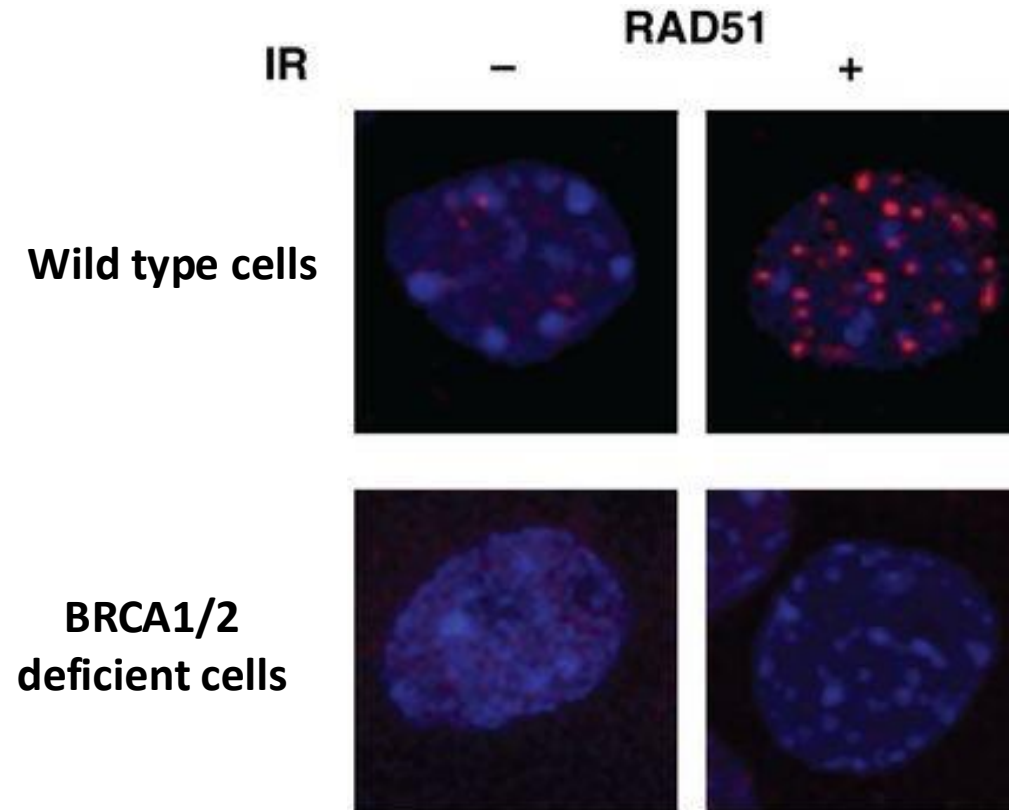
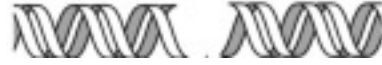
+

Wild type cells

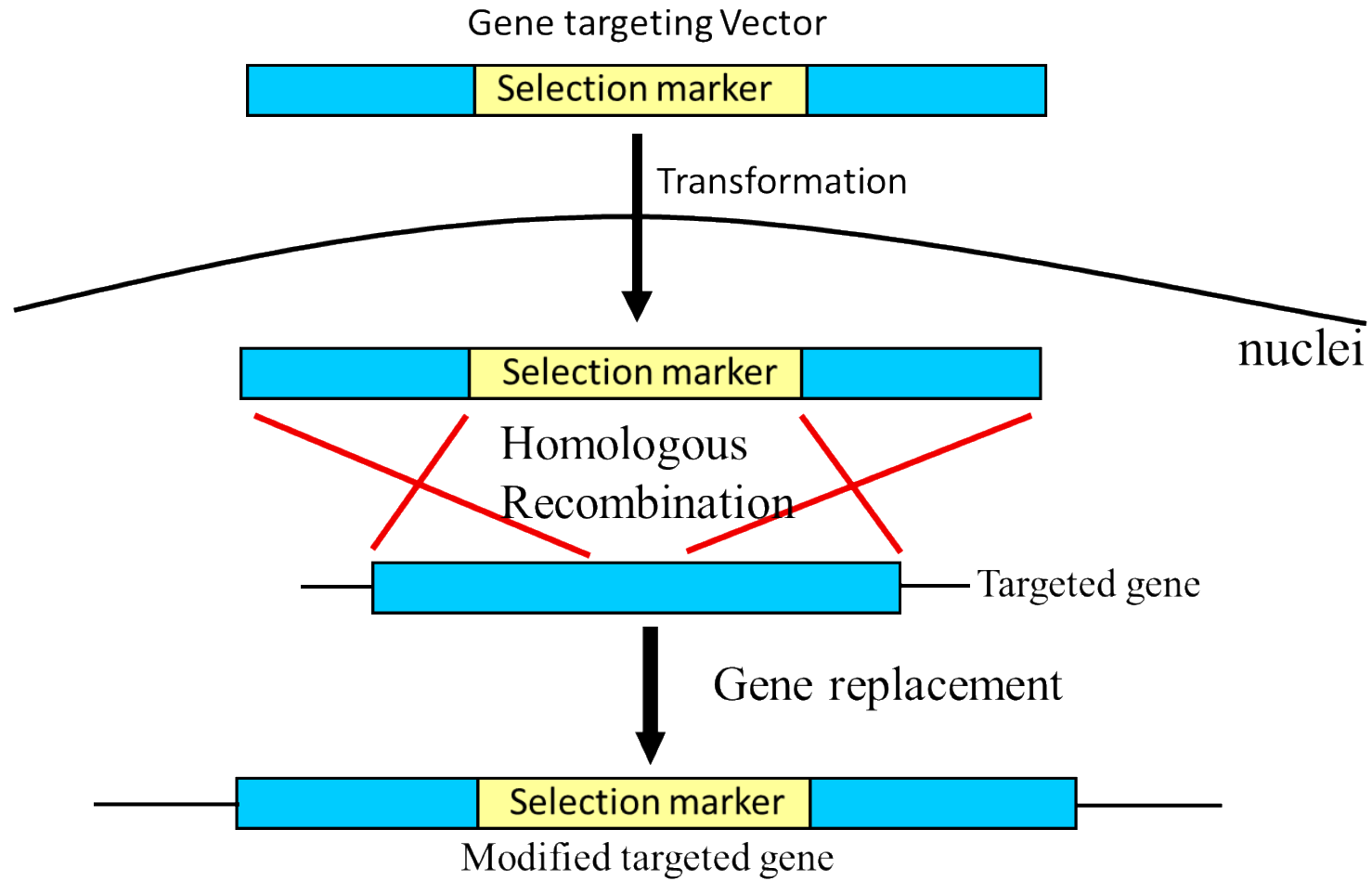


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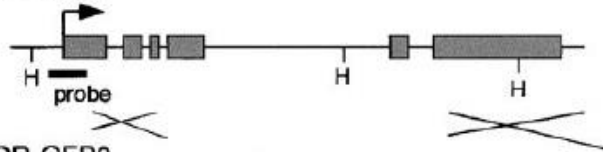


Gene targeting efficiency



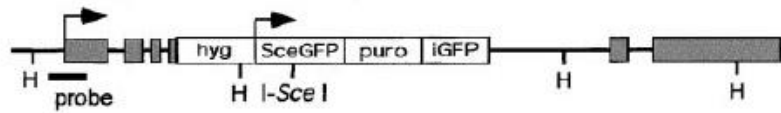
Gene targeting efficiency

A *pim-1* locus

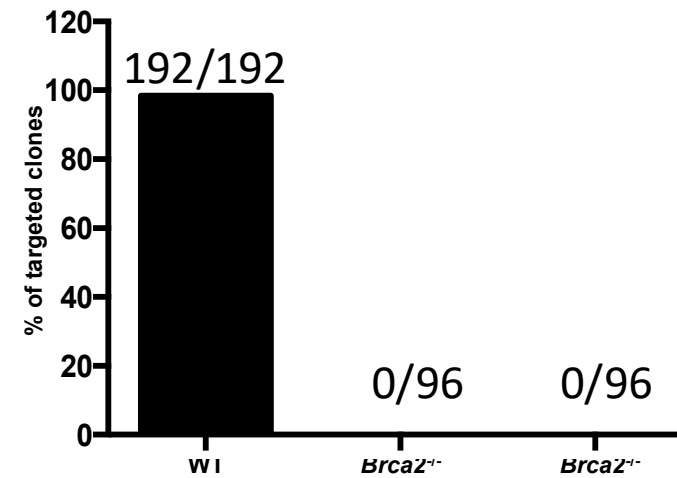
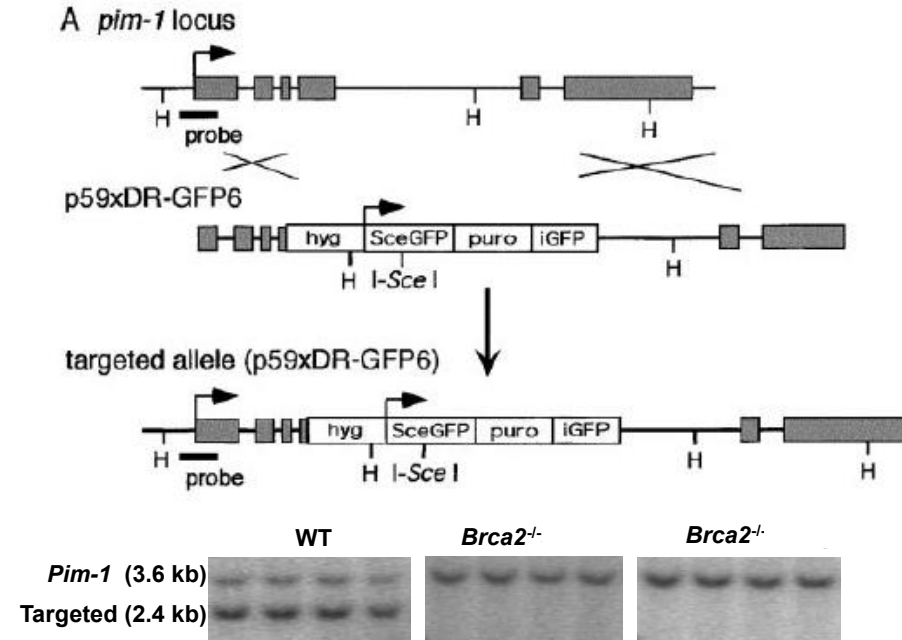


p59xDR-GFP6

targeted allele (p59xDR-GFP6)

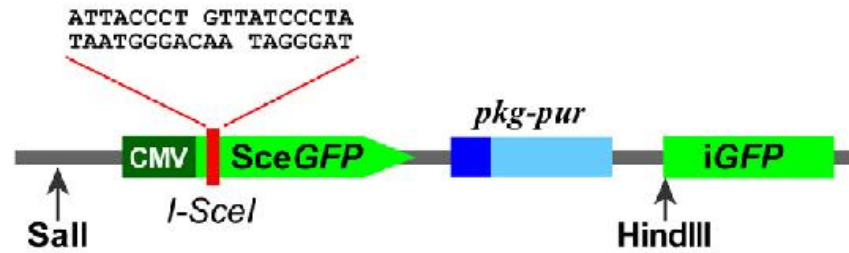


Gene targeting efficiency



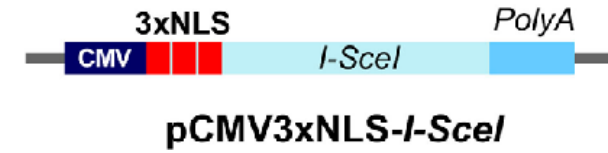
Reporter Assays

DR-GFP



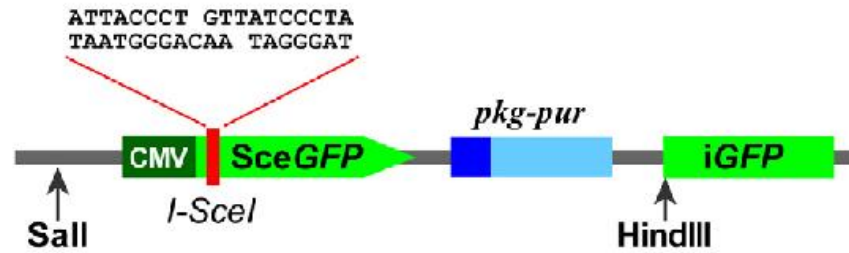
I-SceI recognition sequence

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... 3' -ATCCCTATTGTCCCATTA-5' ...



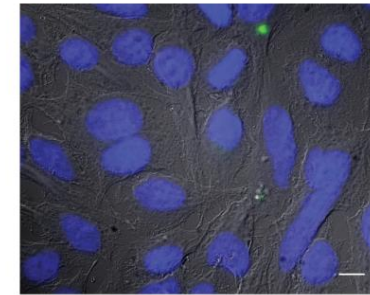
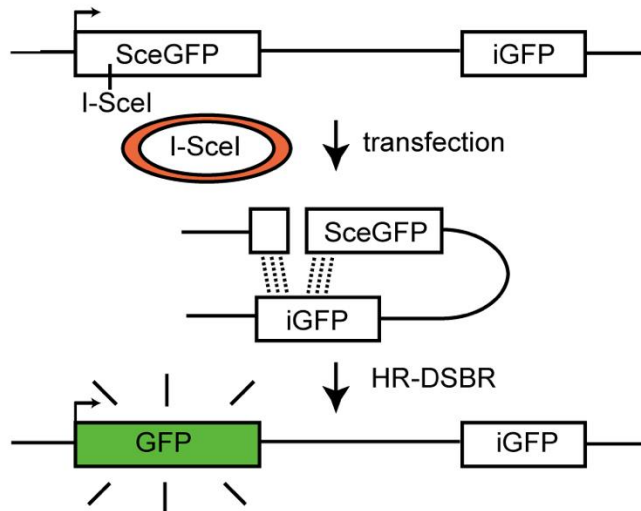
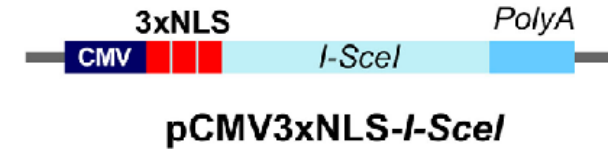
Reporter Assays

DR-GFP

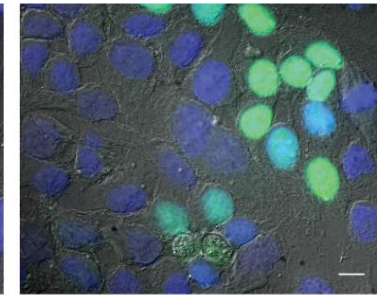


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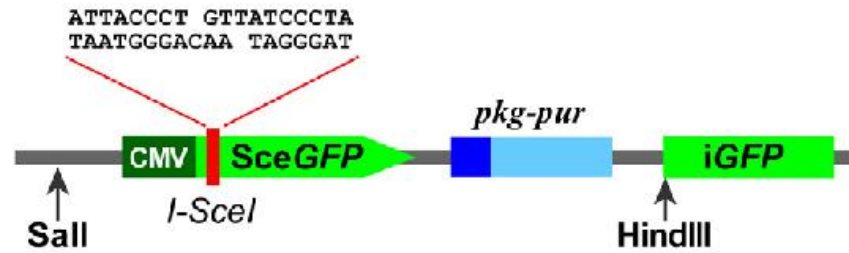
-I-SCEI



+I-SCEI

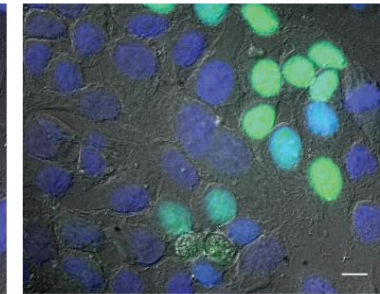
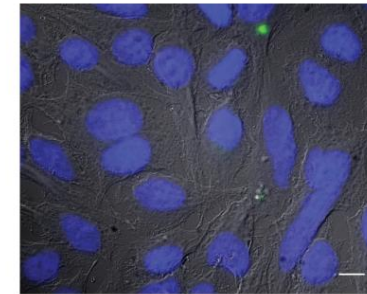
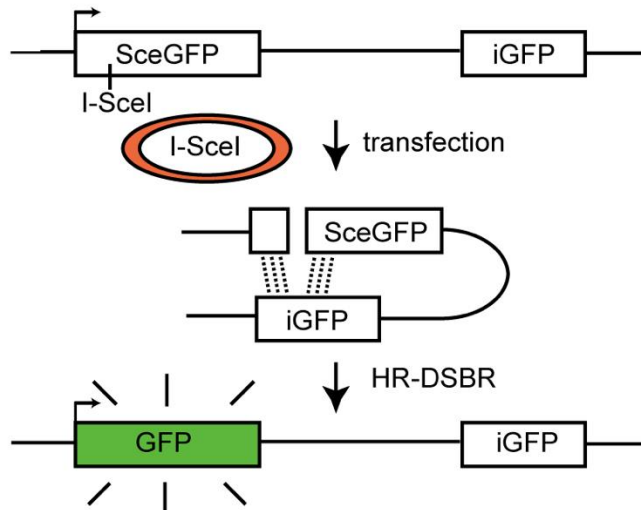
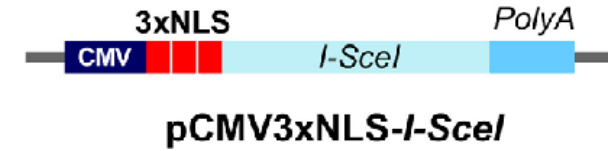
Reporter Assays

DR-GFP

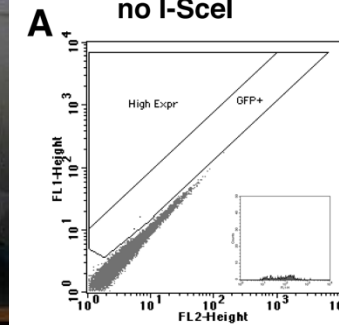


I-SceI recognition sequence

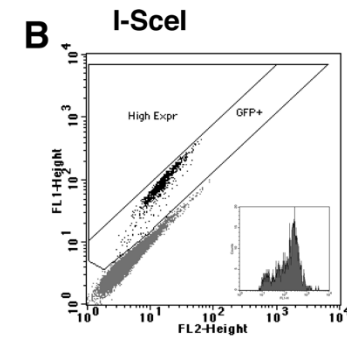
... 5' -TAGGGATAACAGGGTAAT-3' ...
... 3' -ATCCCTATTGTCCCATTA-5' ...



-I-SCEI
no I-SceI

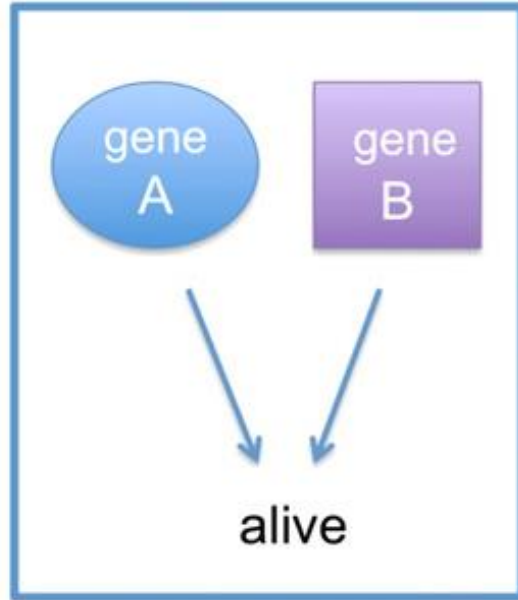


+I-SCEI

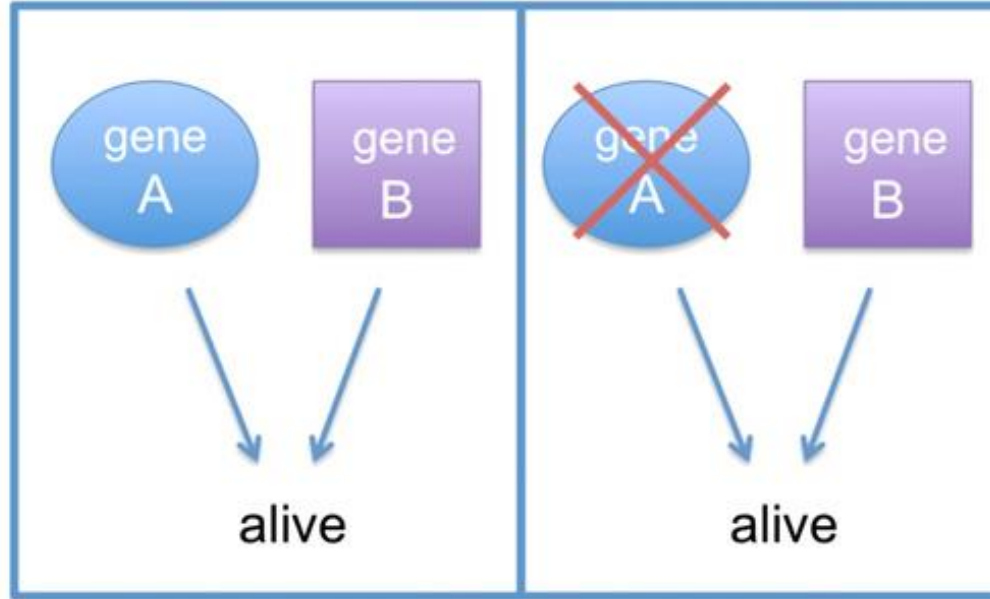


**Can Homologous recombination and replication defects be
exploited for cancer therapy?**

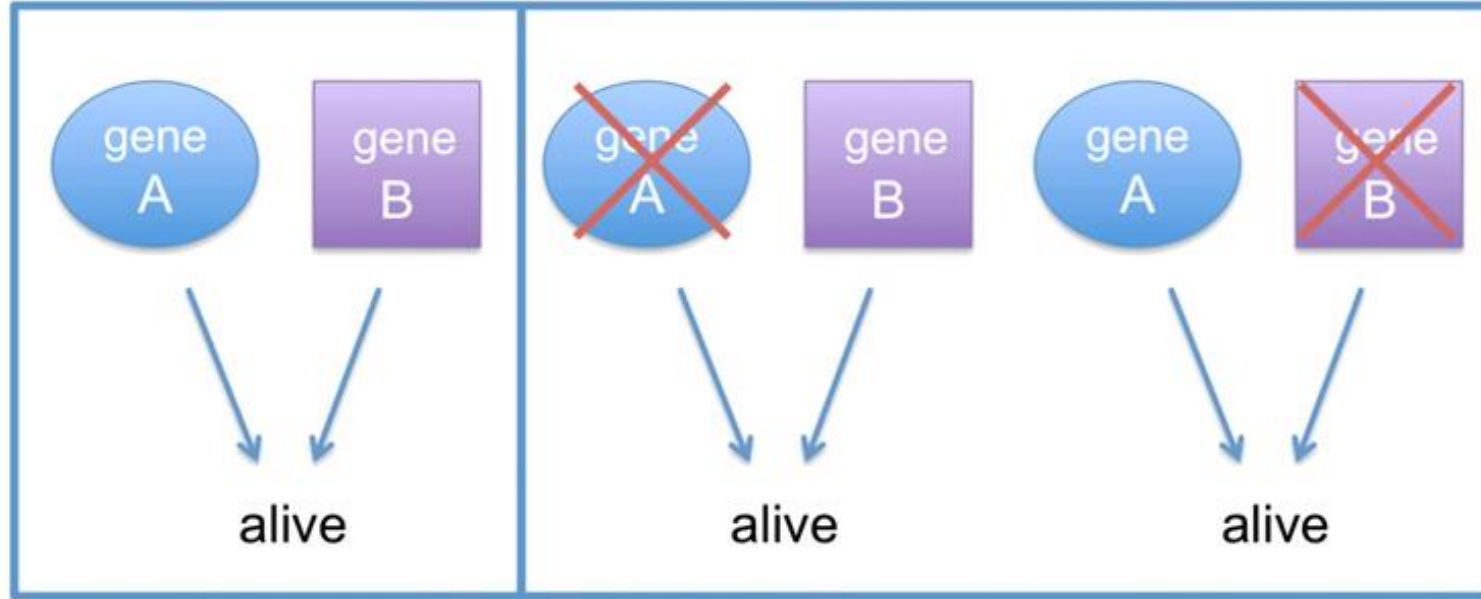
Concept of Synthetic lethality



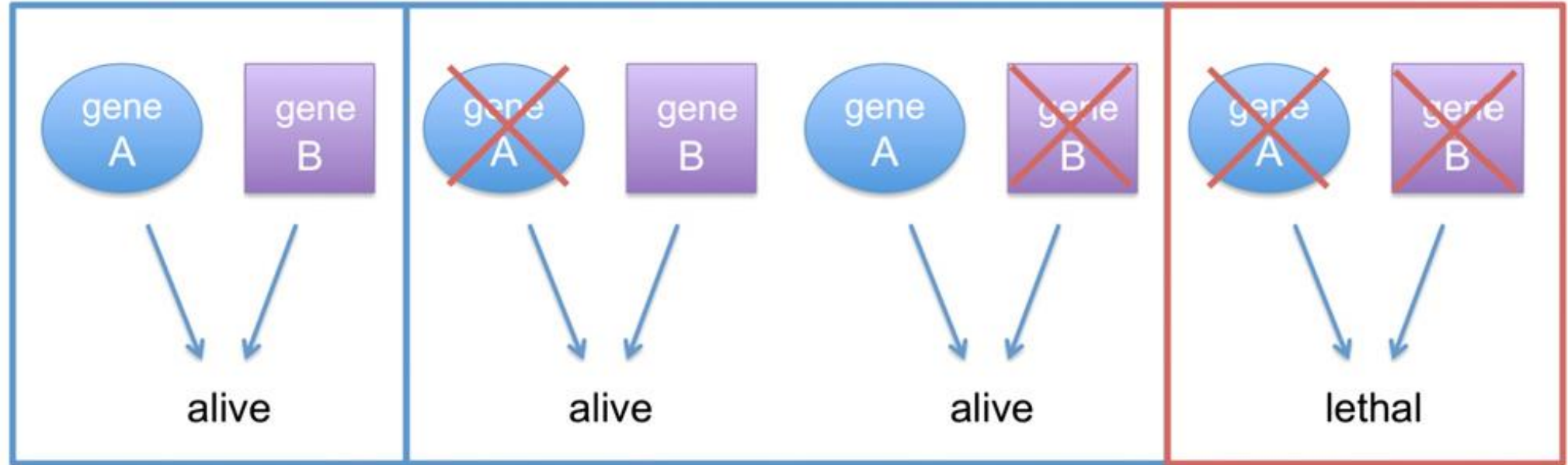
Concept of Synthetic lethality



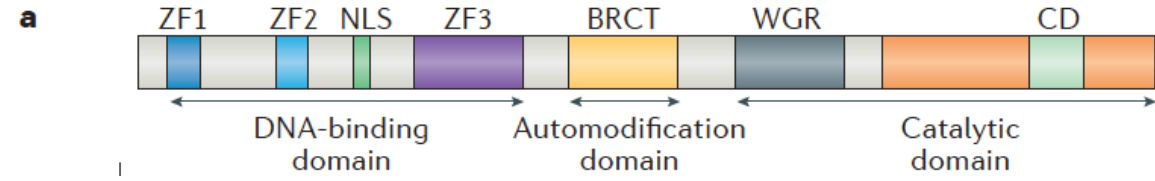
Concept of Synthetic lethality



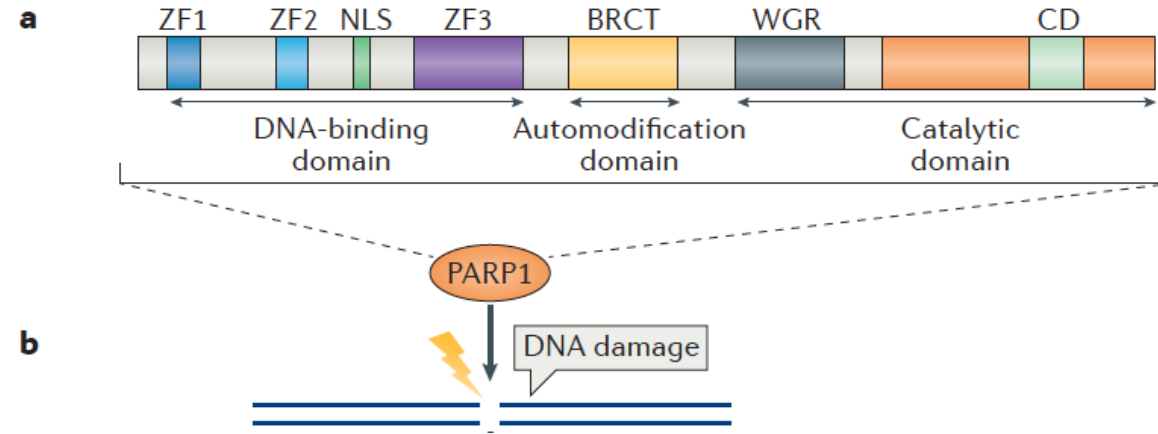
Concept of Synthetic lethality



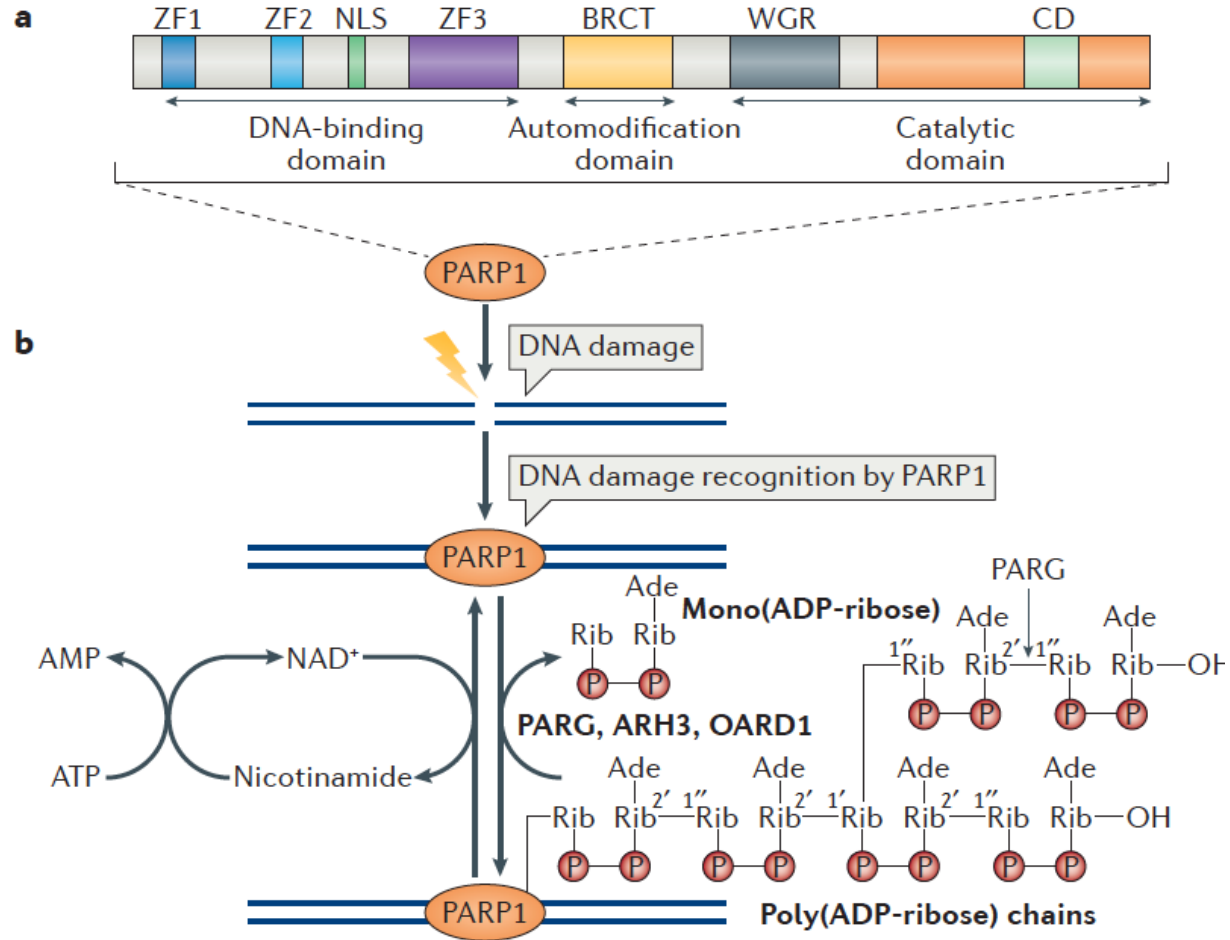
PARP1 in DNA damage



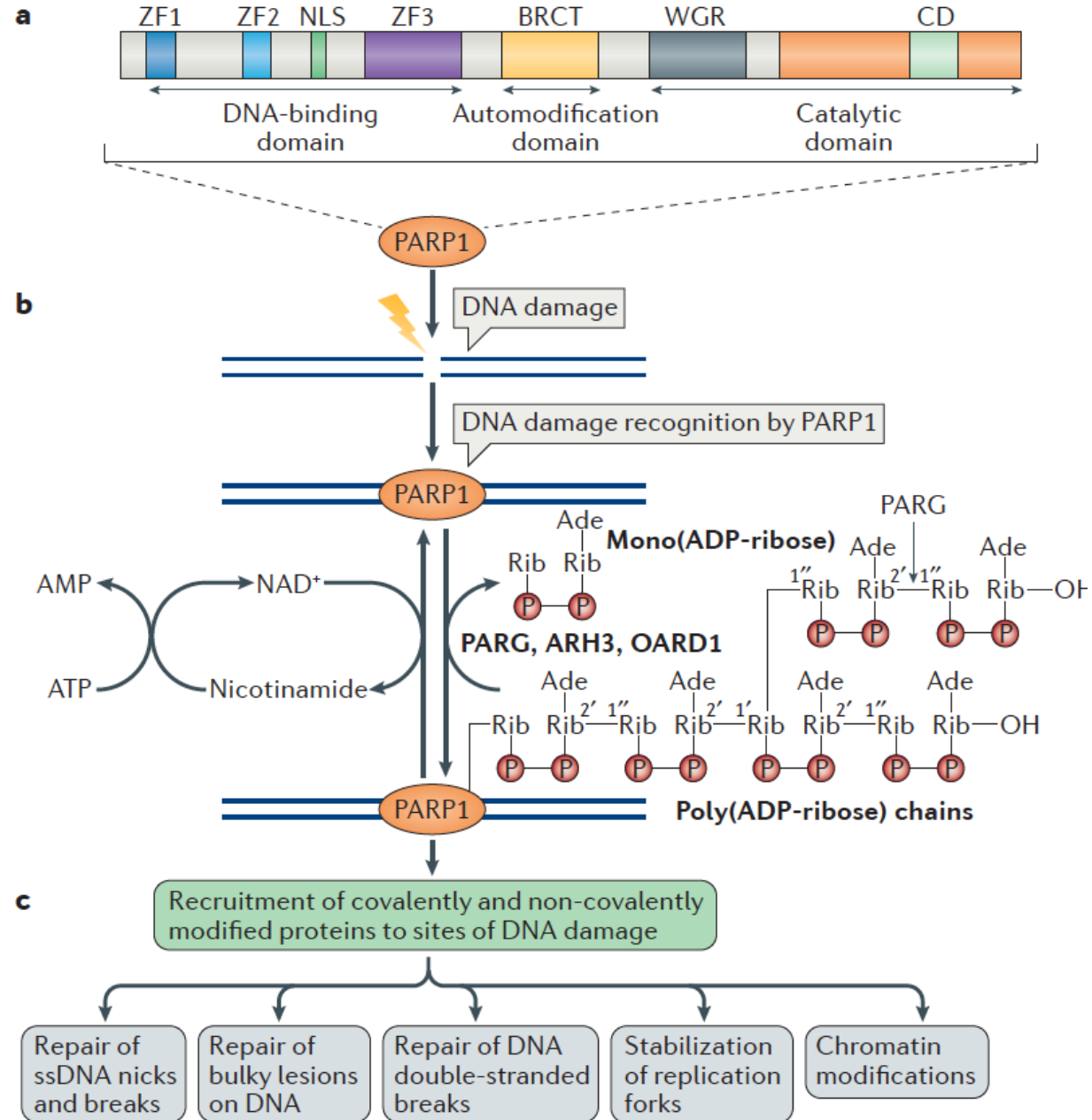
PARP1 in DNA damage



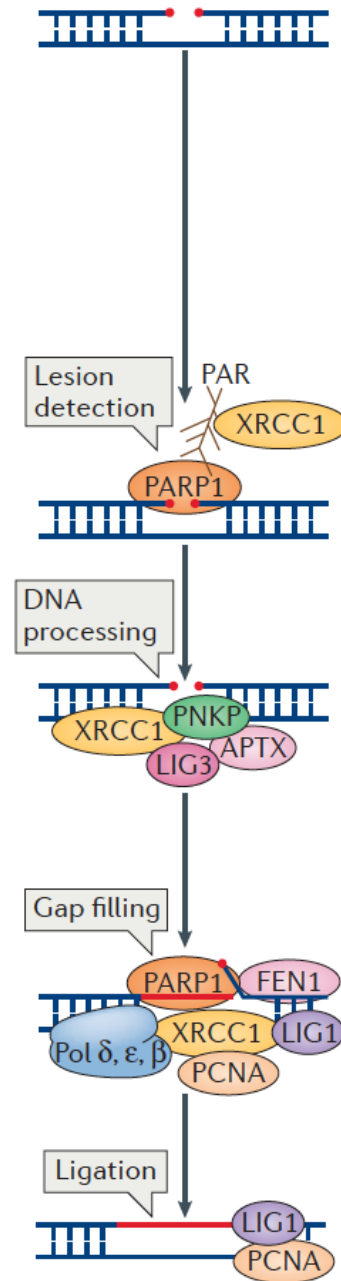
PARP1 in DNA damage



PARP1 in DNA damage

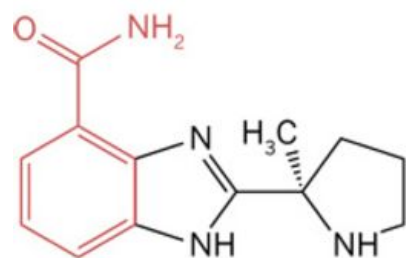


PARP1 in repair of ssDNA breaks

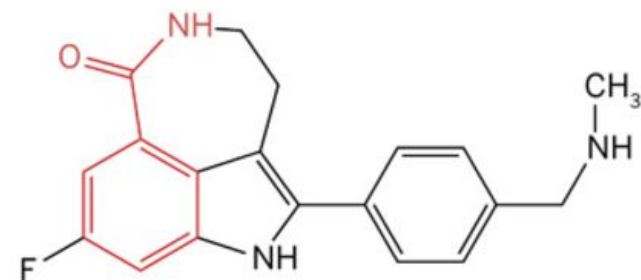


Can **SSB repair** defect and **HR defect** be combined in a **synthetic lethal** approach?

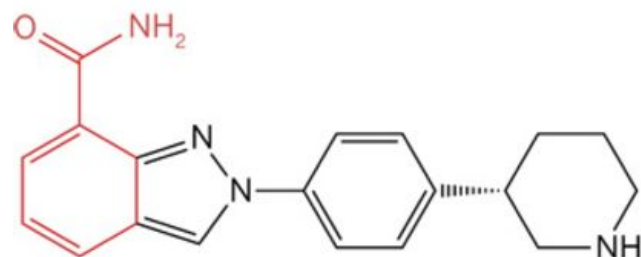
PARP inhibition



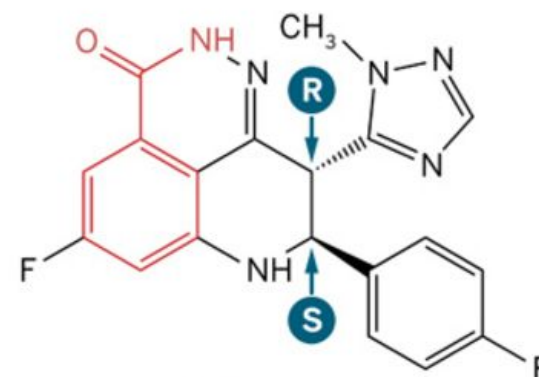
Veliparib
ABT-888



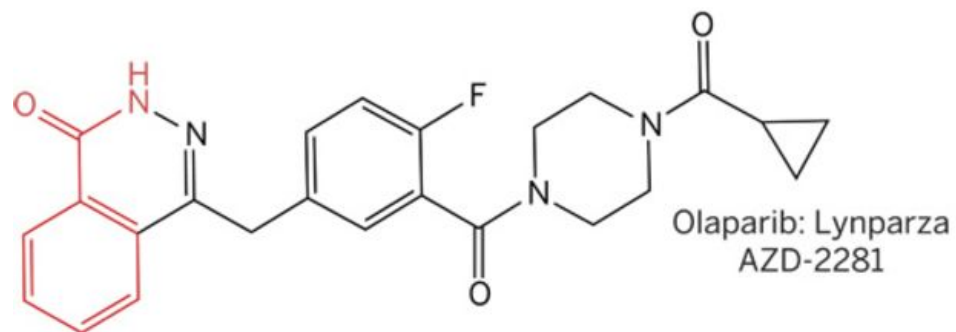
Rucaparib
AG-014699



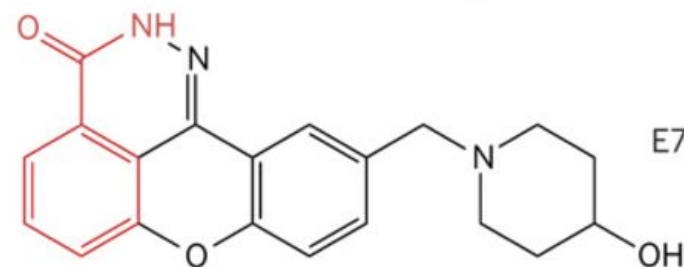
Niraparib
MK-4827



Talazoparib
BMN-673

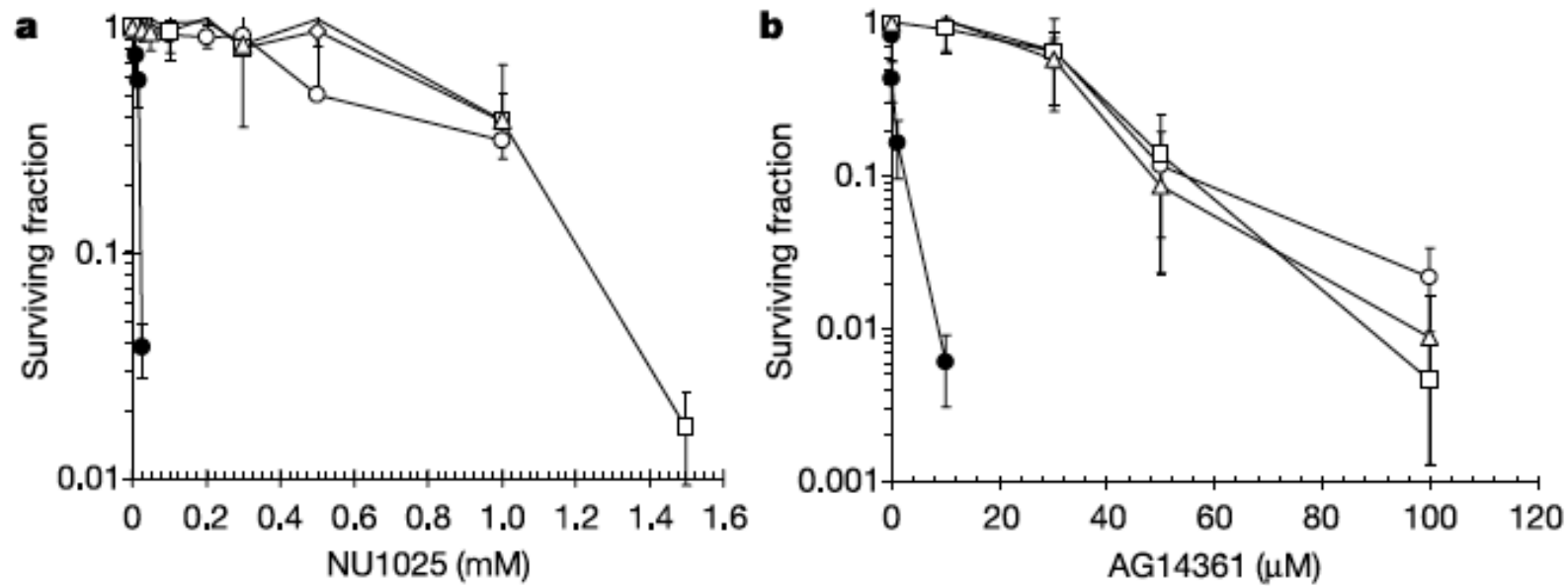


Olaparib: Lynparza
AZD-2281



E7016

BRCA deficient cells are sensitive to PARP inhibition



HOW?

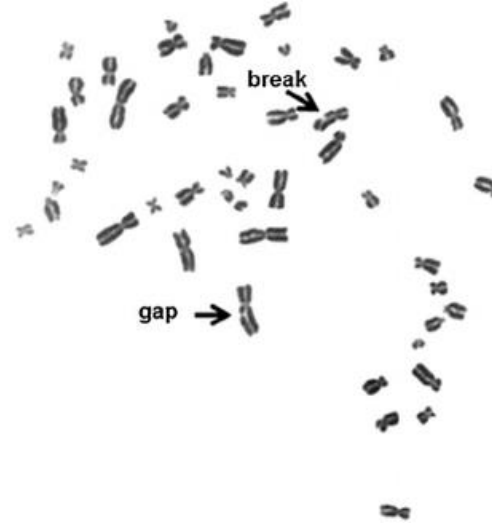
DNA damage?

Metaphase analysis for DNA damage

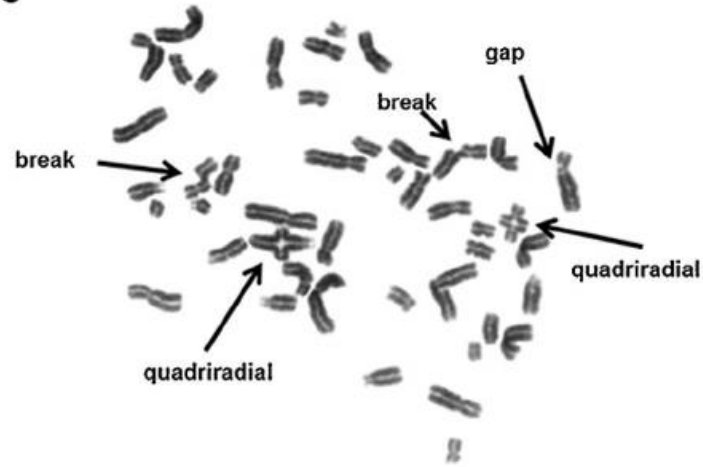
A



B



C

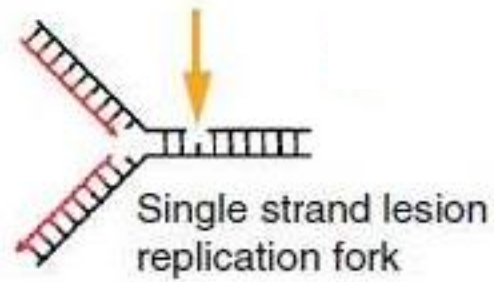


D

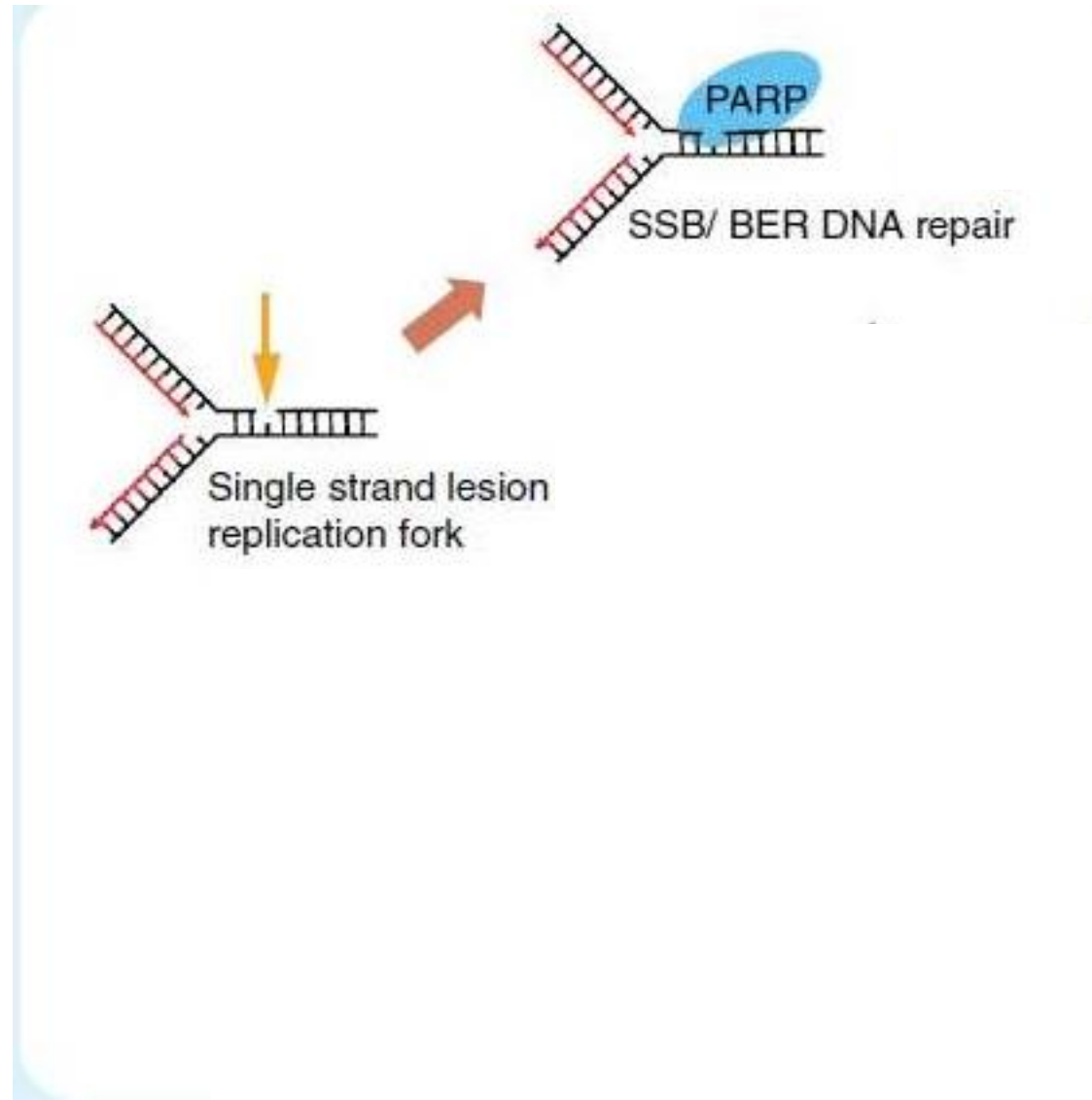
multiple complex damages



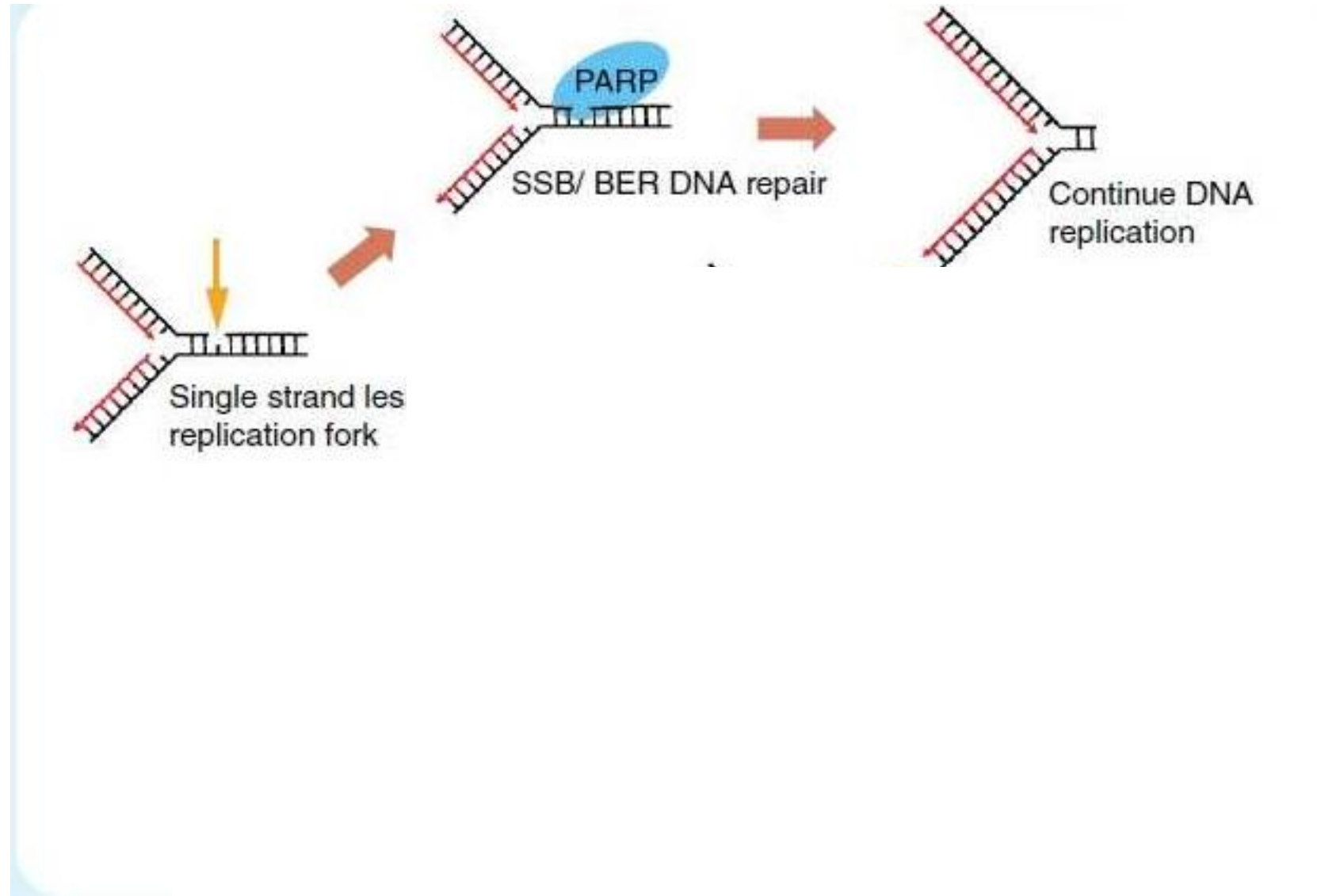
Mechanism for Synthetic lethality between BRCA and PARP deficiency



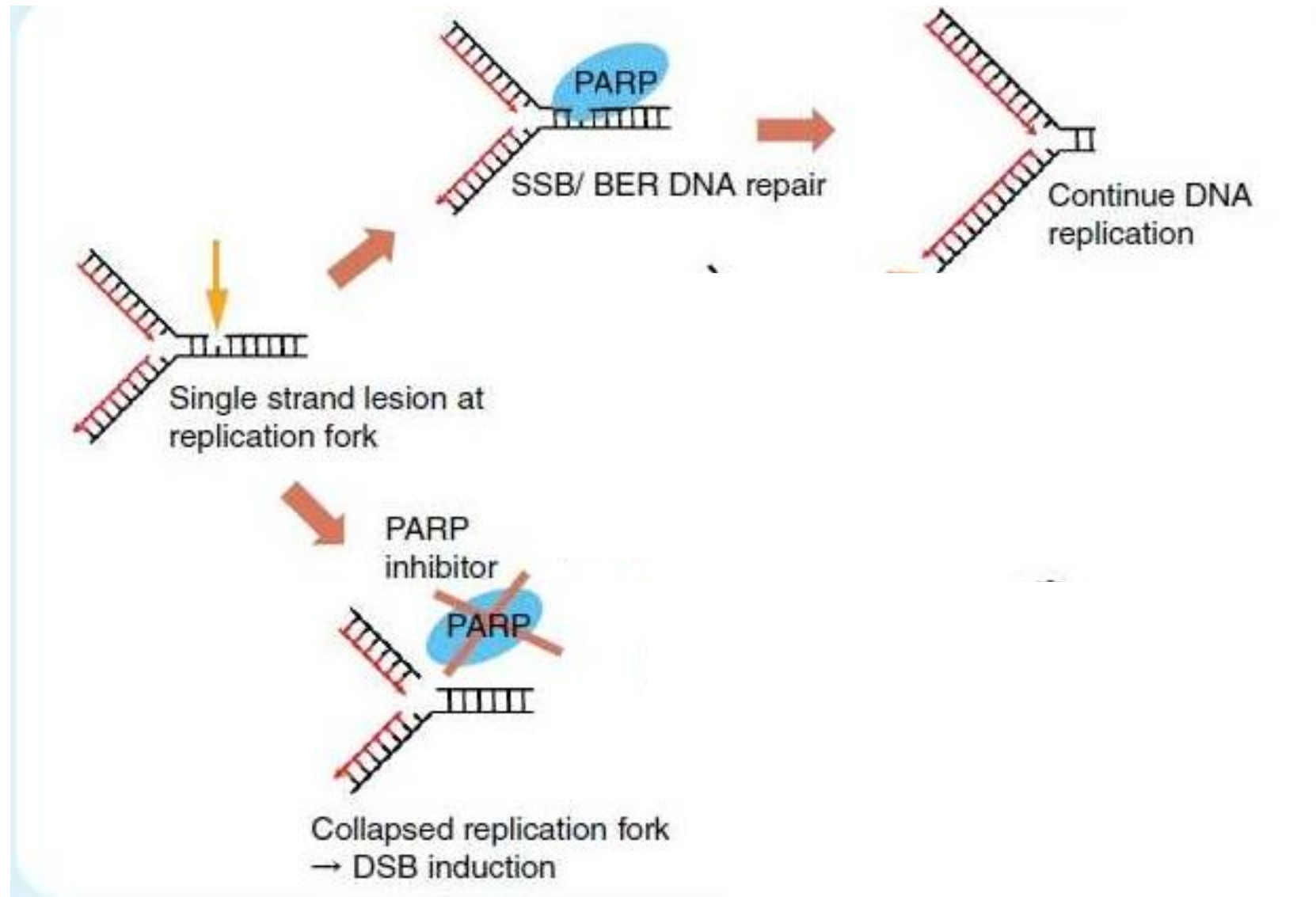
Mechanism for Synthetic lethality between BRCA and PARP deficiency



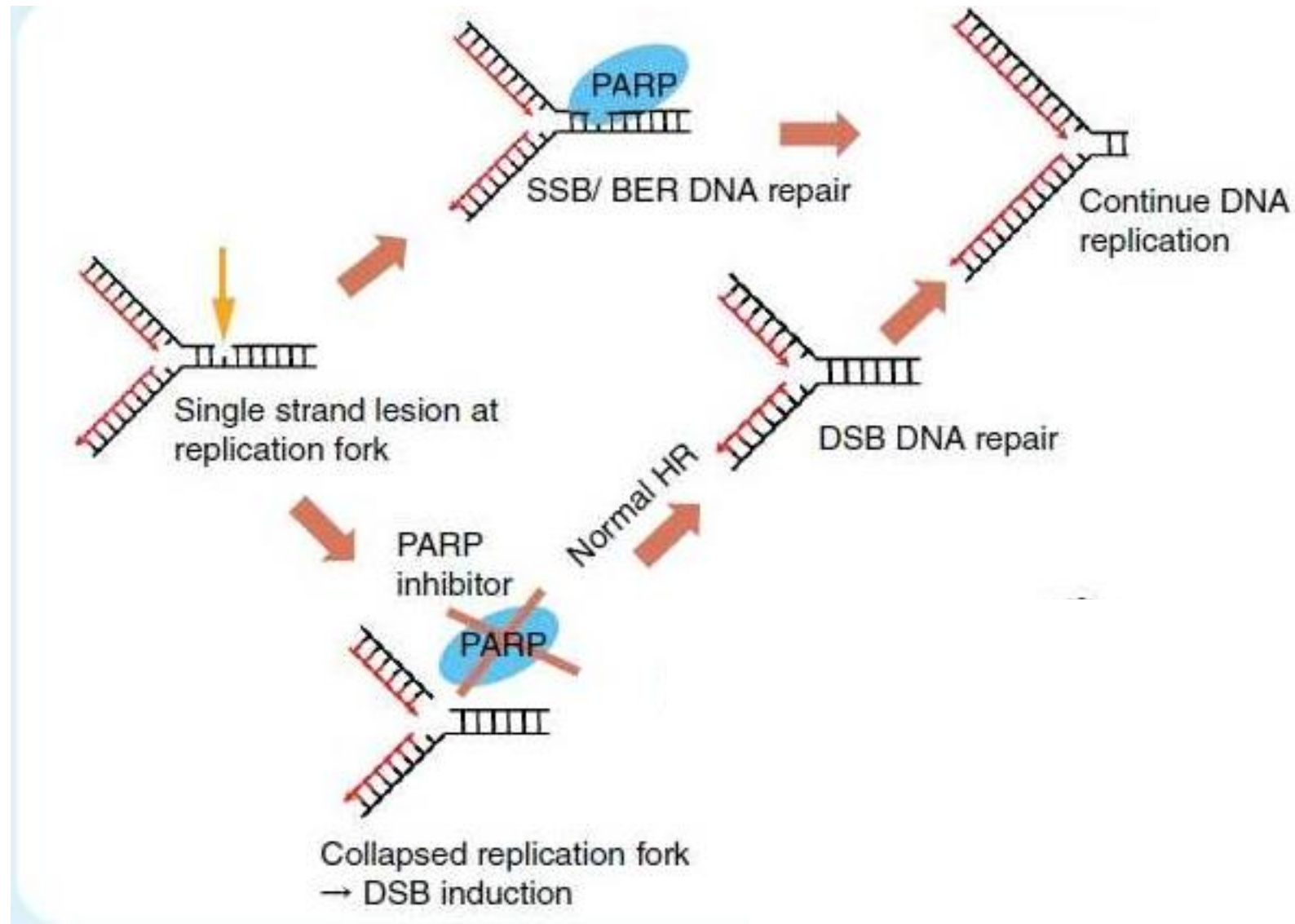
Mechanism for Synthetic lethality between BRCA and PARP deficiency



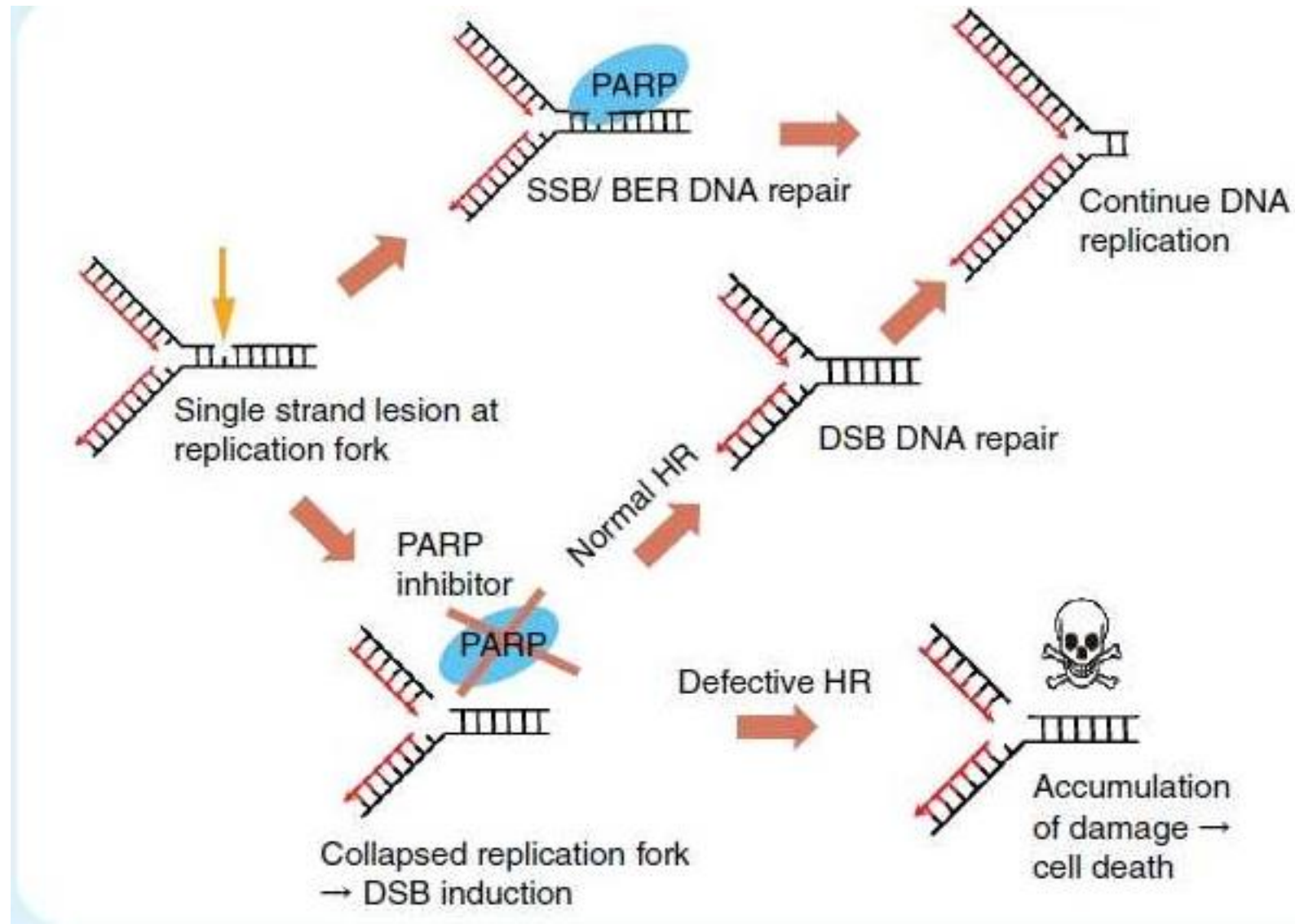
Mechanism for Synthetic lethality between BRCA and PARP deficiency



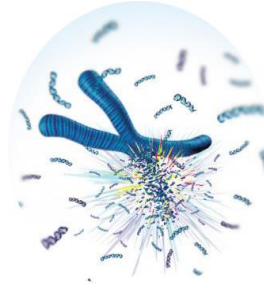
Mechanism for Synthetic lethality between BRCA and PARP deficiency



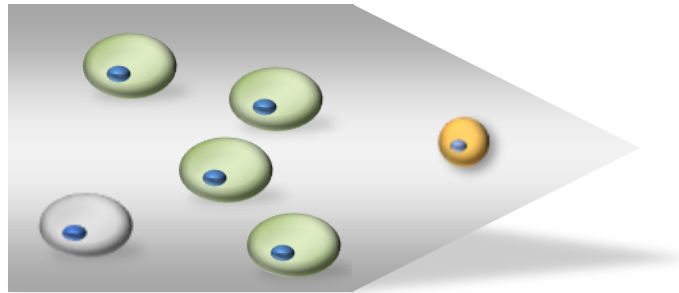
Mechanism for Synthetic lethality between BRCA and PARP deficiency



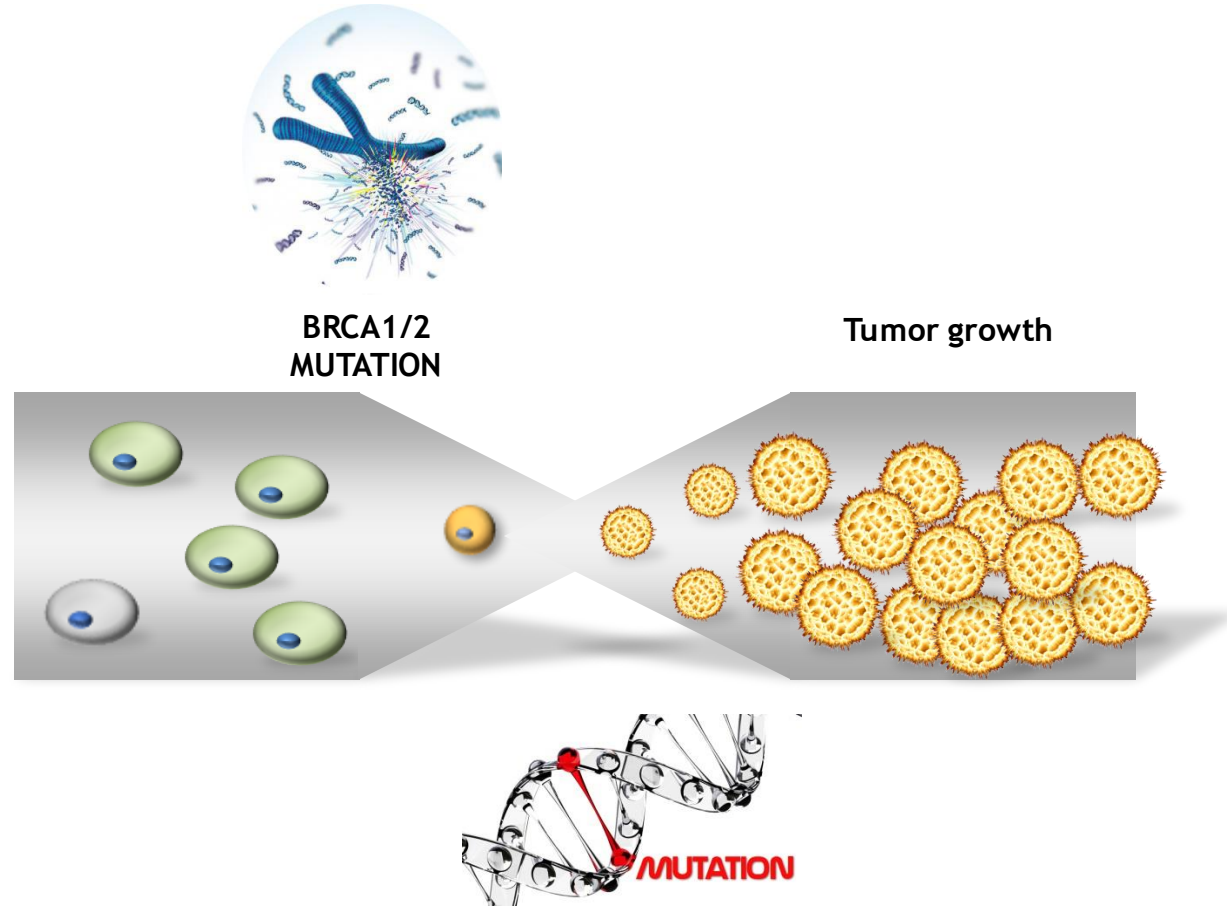
Homologous recombination and replication stress in health and tumorigenesis



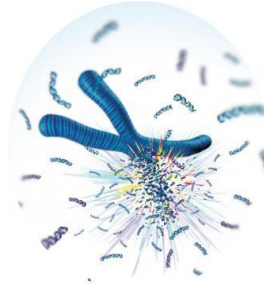
BRCA1/2
MUTATION



Homologous recombination and replication stress in health and tumorigenesis



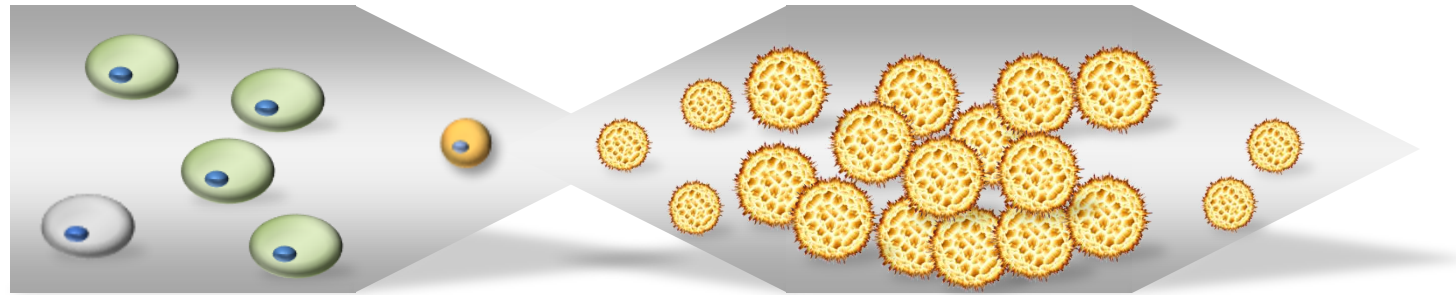
Homologous recombination and replication stress in health and tumorigenesis



BRCA1/2
MUTATION



PARPi/Cisplatin
treatment



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Publications:

1. Kuthethur et al. Ray Chaudhuri*; Science 2025 (in press)
2. Paes/Tripathi et al. Ray Chaudhuri*, 2021; Molecular Cell
3. Mukherjee et al. Ray Chaudhuri*, 2019; Nature Communications
4. Ray Chaudhuri* and Nussenzweig*, 2017; Nature Reviews Molecular Cell Biology (*Corresponding author)
5. Ray Chaudhuri* and Nussenzweig*, 2017; EMBO Molecular Medicine (*Corresponding author)
6. Ray Chaudhuri et al., 2016; Nature
7. Ding, Ray Chaudhuri et al. 2016; Nature Communications
8. Berti, Ray Chaudhuri et al., 2013; Nature Structure and Molecular Biology
9. Ray Chaudhuri et al, 2012; Nature Structure and Molecular Biology

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A M M O D O
S C I E N C E
A W A R D

